BLOCK OPERATOR DECOUPLING IN HILBERT SPACE: INVERSE-FREE SCHUR COMPLEMENTATION AND GAUSSIAN FINITE PARTS

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Abstract. We study the block-operator obstruction behind Schur complementation when the free covariance block is compact and trace class. In this regime the finite-dimensional expression $A - BD^{-1}B^*$ is not an operator formula, because D has no bounded ambient inverse. We replace the inverse by the Anderson–Trapp variational shorting formula on the Cameron–Martin range, derive the canonical λ^{-2} target-block scale from block Lyapunov normalization, prove an $O(\lambda^{-6})$ shorted-residual estimate, and extend the construction to finite-rank targets by Rayleigh–Ritz restriction of the same variational supremum. Gaussian conditioning is then an application of this operator algebra: Galerkin monotonicity follows from subspace nesting, and the renormalized finite part is reduced by Fredholm determinant and trace identities to the invariant quadratic form

$$\frac{1}{2}\delta_E^\top \widetilde{\Sigma}_{EE}^{-1}\delta_E.$$

1. Introduction

The starting point is a functional-analytic obstruction, not a probabilistic one. In finite dimensions, the Schur complement of a positive block matrix is written as $A - BD^{-1}B^*$. For covariance operators on a separable Hilbert space, the free block D is typically compact and trace class; it therefore has no bounded inverse on the ambient space [2, 19, 8]. The finite-dimensional Schur formula is then no longer an operator identity. This is the algebraic obstruction addressed here. A systematic impossibility result (Theorem 11) shows that the obstruction is not merely algebraic: even finite-dimensional truncations suffer condition-number catastrophe $\kappa \sim N^2$, and even with exact arithmetic the asymptotic regime may require $\lambda \gg N^2$, placing the true limit beyond the reach of any fixed truncation. The variational shorting operator is therefore not an elegant alternative—it is the only consistent survival path.

The replacement is Anderson–Trapp shorting. Classical Schur-complement and shorted-operator theory, originating in Krein's theory of non-negative extensions and developed systematically by Anderson and Trapp, gives a

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variational formula for positive block operators [15, 1, 9]. In the covariance setting, its admissibility condition is exactly a Cameron–Martin range condition: cross-covariance vectors must have finite covariance energy rather than lie in the domain of a non-existent bounded inverse. This paper uses that range condition to define the Schur residual as an energy projection, proves its λ^{-6} scale, and shows that Galerkin approximation is only Rayleigh–Ritz restriction of the same supremum to nested finite-dimensional test spaces.

Gaussian conditioning enters after this block-operator decoupling is in place. Singular coordinate conditioning reaches the Feldman–Hájek measure-class boundary: exact evidence is supported on a hyperplane of prior measure zero, while positive-noise Gaussian inverse-problem formulas may remain in the equivalent-measure regime [10, 12, 2, 21, 4, 5]. The block algebra separates the finite shorted covariance residual from the diverging normalization terms. Fredholm determinant identities and trace-anomaly cancellation then give a finite Gaussian Fredholm functional.

Thus the paper's logical order is: block operator decoupling first, Gaussian disintegration second. The stochastic terminology is an application layer. The structural object is inverse-free Schur complementation for compact covariance blocks.

In the rank-one case, the residual has the exact asymptotic form

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad R_\lambda = O(\lambda^{-6}),$$

hence

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}).$$

Section 6 then replaces the scalar target block by a finite matrix $\Sigma_{\lambda,EE} = \lambda^{-2}\widetilde{\Sigma}_{EE}$, proves $\|R_{\lambda,E}\| = O(\lambda^{-6})$, and obtains the normalized matrix coupling $\rho_{\lambda,E} = O(\lambda^{-4})$. These are the powers used in the finite-part cancellation.

The paper has four main contributions:

(1) **Block Lyapunov normalization.** We identify the operator mechanism that fixes the target block at the unique finite-part scale $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$.
(2) **Inverse-free Schur complementation.** We replace the failed free-block inverse by the Anderson–Trapp variational shorting formula on the Cameron–Martin range, and prove the $O(\lambda^{-6})$ shorted-residual estimate.
(3) **Galerkin monotonicity.** We prove that finite-rank Galerkin truncations converge monotonically by expanding the variational supremum. Loewner monotonicity is a consequence of nested subspaces, not matrix inversion.
(4) **Trace-anomaly cancellation.** For Gaussian Radon laws, disintegration supplies the exact-evidence measure, while Weinstein–Aronszajn

and Fredholm determinant identities cancel the trace anomalies and reduce the finite part to the invariant quadratic form.

1.1. **Positioning in the literature.** The paper is closest to four bodies of work. First, it uses Schur complements, shorted operators, Douglas range inclusion, and modern complementability theory for block operators [15, 1, 9, 22, 7, 11, 13]. Second, it uses the measure-class structure of Gaussian measures, especially the Cameron–Martin theorem and Feldman–Hájek dichotomy [10, 12, 2]. Third, it is adjacent to Bayesian inverse problems and Hilbert-space Gaussian conditioning, including shorted covariance descriptions, positive noise inverse-problem formulas, low-rank covariance approximations, and nonlinear observation maps [18, 21, 6, 20, 4, 5]. Fourth, the examples come from stochastic evolution equations and elliptic Gaussian fields, where dissipative sectorial generators produce compact trace-class stationary covariances naturally approximated by Galerkin truncation [8, 17, 19].

1.2. **Organization of the paper.** Section 2 defines the block operator model and canonical normalization. Section 3 establishes inverse-free Hilbert-space Schur complementation. Section 4 applies the operator result to Radon Gaussian disintegration and Fredholm finite parts. Section 5 gives an elliptic Gaussian-field example. Section 6 develops finite-rank shorting, operator-valued residuals, misalignment, and exact matrix/Fredholm cancellation. Section 7 concludes. Proofs are given in Sections A and B; numerical diagnostics and Galerkin verification are given in Section C.

2. Operator Model and Canonical Normalization

2.1. **Operator Evolution and Regularized Conditioning.** We model the continuous-time process as stochastic evolution on a separable Hilbert space $\mathcal{H}$ [8]:

$$dX_t = \mathcal{A}X_t\,dt + \mathcal{D}^{1/2}dW_t \tag{1}$$

where $\mathcal{A}$ is the drift operator and $\mathcal{D}$ is the trace-class diffusion operator. We decompose the Hilbert space as $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$, where $\mathcal{H}_k := \mathbb{R}\phi_k$ is the one-dimensional target coordinate subspace, and $\mathcal{H}_I := \mathcal{H}_k^\perp$ is the free bulk subspace. Let $\Pi_k := \phi_k \otimes \phi_k$ and $\Pi_I := I - \Pi_k$ be the corresponding orthogonal projections.

To approximate the exact point conditioning $X_k = x_k^*$, we set the incoming target-coordinate blocks to zero and apply a regularization parameter $\lambda > 0$.

Definition 1 (Calibrated Approximating Operator Family)**.** The regularized conditioning operator of strength $\lambda > 0$ at coordinate X_k acts on the operator evolution (1) by modifying both the drift and the diffusion:

$$\mathcal{A}^{(\lambda)} := \mathcal{A}_{\mathrm{cond}} - \lambda\Pi_k, \qquad \mathcal{D}^{(\lambda)} := \mathcal{D}_{\mathrm{cond}} + d_\lambda\Pi_k \tag{2}$$

where $\mathcal{A}_{\mathrm{cond}}$ is the block drift operator with incoming target-coordinate blocks set to zero ($\Pi_k \mathcal{A}_{\mathrm{cond}} = 0$), $\mathcal{D}_{\mathrm{cond}}$ has target-coordinate cross-noise set to zero, and $d_\lambda > 0$ is the local diffusion variance of the target coordinate at regularization strength λ. In the SDE, this corresponds to:

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{d_\lambda}\, dW_{k,t}. \tag{3}$$

2.2. **Canonical Normalization.** The block-decoupled structure under the regularized operator family yields:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} d_\lambda & 0 \\ 0 & D_{II} \end{pmatrix}, \tag{4}$$

where $a_{Ik} := \mathcal{A}_{Ik}\phi_k \in \mathcal{H}_I$ represents the off-diagonal drift coupling. Let Σ_λ denote the stationary covariance operator under regularization strength λ, solving the Lyapunov equation:

$$\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0. \tag{5}$$

The target block kk decouples completely from the free coordinates, yielding the exact scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + d_\lambda = 0, \qquad \Sigma_{\lambda,kk} = \frac{d_\lambda}{2\lambda}. \tag{6}$$

The normalization is fixed by canonical scaling, not by a free modeling convention. In the Gaussian finite-part extension below, a constant forcing b_k produces the deterministic target offset

$$\Delta_\lambda := m_{\lambda,k} - x_k^* = \frac{b_k}{\lambda} = O(\lambda^{-1}), \tag{7}$$

so the squared deterministic displacement has the rigid scale $\Delta_\lambda^2 = b_k^2/\lambda^2 = O(\lambda^{-2})$. The leading target contribution to the conditional finite-part functional is the signal-to-noise ratio

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2/\lambda^2}{2\Sigma_{\lambda,kk}}. \tag{8}$$

Consequently the target covariance must decay at the same order as the squared deterministic displacement. If $\Sigma_{\lambda,kk} = O(\lambda^{-1})$, as for the slow family $d_\lambda = \varepsilon_0$, then (8) is $O(\lambda^{-1})$ and the finite target contribution collapses to zero. If $\Sigma_{\lambda,kk} = O(\lambda^{-3})$, as for the fast family $d_\lambda = \varepsilon_0/\lambda^2$, then (8) is $O(\lambda)$ and the target contribution diverges. These two failures are the slow and fast ablations reported in Section C.2. A nonzero finite limit therefore forces the critical covariance scale

$$\Sigma_{\lambda,kk} \asymp \lambda^{-2}. \tag{9}$$

Proposition 2 (Canonical Scaling)**.** *The unique target-diffusion scaling that makes the deterministic offset and target variance balance at a finite nonzero scale is*

$$\Sigma_{\lambda,kk} \equiv \frac{\varepsilon_0}{2\lambda^2}, \qquad \varepsilon_0 > 0. \tag{10}$$

By the scalar Lyapunov identity (6)*, this is equivalent to*

$$d_\lambda = 2\lambda\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{\lambda} \quad \textit{for every } \lambda > 0. \tag{11}$$

With this canonical family,

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2}{\varepsilon_0}, \tag{12}$$

and the calibrated approximating operators become

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{\varepsilon_0/\lambda}\, dW_{k,t}. \tag{13}$$

All subsequent results are stated for this canonical scaling family.

Definition 3 (Shorted Schur Energy Invariant)**.** Let $S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I} X_k\|_{L^2}^2$ represent the shorted Schur projection residual of the target coordinate X_k against the closed linear span $\mathcal{M}_I$ of the free coordinates X_I. In probabilistic language this is the linear minimum mean squared error residual, but the operator object is the Schur energy projection. The *shorted Schur invariant* m_2^{Schur} is defined as:

$$m_2^{\text{Schur}} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}. \tag{14}$$

2.3. **Operator Setup and Assumptions.** We state the operator-theoretic assumptions that make the limit passage independent of the Galerkin dimension.

Definition 4 (Drift Operator with Analytic Base)**.** A drift operator on a separable Hilbert space $\mathcal{H}$ is an operator of the form $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where:

- (i) $\mathcal{A}_0 : \mathrm{dom}(\mathcal{A}_0) \subset \mathcal{H} \to \mathcal{H}$ is closed, densely defined, and dissipative, and it generates a strongly continuous analytic contraction semigroup $(e^{t\mathcal{A}_0})_{t\geq 0}$ on $\mathcal{H}$ satisfying $\|e^{t\mathcal{A}_0}\| \leq e^{-\alpha t}$ for some spectral gap $\alpha > 0$; equivalently, the base operator $-\mathcal{A}_0$ is sectorial of angle $< \pi/2$ in this exponentially stable contraction setting.
- (ii) $\mathcal{K} : \mathcal{H} \to \mathcal{H}$ is a bounded linear operator.
- (iii) The full operator $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, with $\mathrm{dom}(\mathcal{A}) = \mathrm{dom}(\mathcal{A}_0)$, generates a strongly continuous analytic semigroup $(e^{t\mathcal{A}})_{t\geq 0}$ by the bounded perturbation theorem, and this semigroup remains exponentially stable: $\|e^{t\mathcal{A}}\| \leq Me^{-\beta t}$ for some $M \geq 1$, $\beta > 0$.

In finite dimensions ($\mathcal{H} = \mathbb{R}^n$), this reduces to a decomposition of a Hurwitz drift matrix into an exponentially stable analytic base plus a bounded perturbation.

Definition 5 (Trace-Class Noise and Covariance)**.** Let $\mathcal{S}_1(\mathcal{H})$ denote the Banach space of trace-class (nuclear) operators on $\mathcal{H}$ equipped with the trace norm $\|T\|_{\mathcal{S}_1} := \mathrm{Tr}(|T|) = \sum_{i=1}^\infty \langle \phi_i, (T^*T)^{1/2}\phi_i\rangle < \infty$.

- (i) The diffusion operator $\mathcal{D} : \mathcal{H} \to \mathcal{H}$ is positive, self-adjoint, and belongs to $\mathcal{S}_1(\mathcal{H})$.

(ii) Assuming the pair $(\mathcal{A}, \mathcal{D}^{1/2})$ is controllable, the stationary covariance operator $\Sigma : \mathcal{H} \to \mathcal{H}$ is the unique strictly positive, self-adjoint, trace-class solution to the operator Lyapunov equation:

$$\mathcal{A}\Sigma + \Sigma\mathcal{A}^* + \mathcal{D} = 0 \tag{15}$$

which is given by the convergent Bochner integral

$$\Sigma = \int_0^\infty e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}dt.$$

Assumption 6 (Core Analytical Assumptions for Singular Hyperplane Conditioning)**.** This work relies on two core analytical assumptions regarding the target-coordinate block decoupling and the outgoing coupling:

(i) **Linear Second-Moment Channel:** A centered linear stationary process on $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ has finite second moments and covariance solving Equation (15). The block-decoupling constraint zeroes all incoming target drift and cross-noise, so that $\mathcal{A}_{kI} = \mathcal{D}_{kI} = \mathcal{D}_{Ik} = 0$ and $\mathcal{A}_{kk} = -\lambda$. The target diffusion $\mathcal{D}_{kk} = \varepsilon_0/\lambda$ is the unique scaling determined by the canonical normalization (Proposition 2). The free noise is independent of the local target noise. (Gaussianity is not assumed).

(ii) **Cameron–Martin Stability of Off-Diagonal Drift Coupling:** Let $Q_0 \coloneqq \Sigma_{II}^{(0)}$ denote the free-block covariance produced by the free noise under block decoupling, and let $H_{Q_0} \coloneqq \mathrm{Range}(Q_0^{1/2})$ with the energy norm $|u|_{Q_0} \coloneqq \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. The outgoing vector $a_{Ik} \coloneqq \mathcal{A}_{Ik}\phi_k$ belongs to H_{Q_0}, and $e^{t\mathcal{A}_{II}}$ restricts to a strongly continuous exponentially stable semigroup on H_{Q_0}:

$$\|e^{t\mathcal{A}_{II}}\|_{\mathcal{L}(H_{Q_0})} \le Me^{-\omega t}, \qquad t \ge 0, \tag{16}$$

for constants $M \ge 1$ and $\omega > 0$ independent of λ.

Intuitive Explanation: Part (i) states that the regularized operator family act as a strong localized drift penalization on the target coordinate X_k while isolating it from incoming influences, giving an operator implementation of the block-decoupling constraint. Part (ii) requires that the off-diagonal drift coupling a_{Ik} from the target coordinate has finite energy relative to the background fluctuations of the free coordinates. This ensures that target fluctuations propagating downstream do not overload the variance of the infinite-dimensional system.

Remark 7 (Finite-Energy Outgoing Covariance Coupling and Finite-Dimensional Calibration). The membership $a_{Ik} \in H_{Q_0}$ is an admissibility condition, not a consequence of the free-block Lyapunov equation alone: an arbitrary outgoing vector need not lie in $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$. It says that the deterministic channel carrying target fluctuations into the free-block has finite covariance energy relative to the free-noise background.

To understand this condition intuitively, consider a finite-dimensional setting where the free-block state space is $\mathcal{H}_I = \mathbb{R}^{d-1}$ and the free stationary covariance $Q_0 \coloneqq \Sigma_{II}^{(0)}$ is strictly positive-definite. Since $Q_0 \succ 0$, its square root $Q_0^{1/2}$ is invertible, and its range is the entire space $\mathbb{R}^{d-1}$. In this case, the Cameron–Martin space is the full space $H_{Q_0} = \mathrm{Range}(Q_0^{1/2}) = \mathbb{R}^{d-1}$, and the Cameron–Martin energy norm $|u|_{Q_0} = \|Q_0^{-1/2}u\|_2$ is equivalent to the standard Euclidean norm. Consequently, the admissibility condition $a_{Ik} \in H_{Q_0}$ is trivially satisfied for any outgoing vector $a_{Ik} \in \mathbb{R}^{d-1}$.

However, if some direction in the free coordinates is noise-free under block decoupling (meaning Q_0 is positive semidefinite but singular), H_{Q_0} becomes a proper subspace of $\mathbb{R}^{d-1}$. In this case, the condition $a_{Ik} \in H_{Q_0}$ requires that the leakage vector a_{Ik} has no component in the null space of Q_0. Equivalently, it prevents the target coordinate from leaking fluctuations into a downstream coordinate that has zero background noise, which would otherwise result in infinite relative covariance energy.

In infinite dimensions, $\Sigma_{II}^{(0)}$ is trace-class and compact, so its eigenvalues decay to zero, making $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$ a dense but strictly proper subspace of $\mathcal{H}_I$. The condition $a_{Ik} \in H_{Q_0}$ requires the off-diagonal drift coupling coefficients to decay sufficiently fast relative to the background noise eigenvalues to keep the relative energy finite.

Under this assumption, invariance and exponential stability on H_{Q_0} imply:

$$
\begin{aligned}
&(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik} \in H_{Q_0}, \\
&|(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}|_{Q_0} \le \frac{M}{\lambda + \omega} |a_{Ik}|_{Q_0}.
\end{aligned}
\tag{17}
$$

Remark 8 (Constant Dependency in Schur Energy Expansion). Consequently, the exact leakage cross-covariance $\Sigma_{\lambda,Ik}$ has finite Cameron–Martin energy norm and is $O(\lambda^{-3})$ in that norm. This is a condition on the deterministic response and covariance energy; it does not assert that infinite-dimensional Gaussian sample paths lie in H_{Q_0} almost surely.

3. Inverse-Free Schur Complementation in Hilbert Space

3.1. **Variational Shorting and Energy Bounds.** In finite dimensions, the Schur complement of the bulk covariance block Σ_{II} is defined as $S_k \coloneqq \Sigma_{kk} - \Sigma_{kI}\Sigma_{II}^{-1}\Sigma_{Ik}$. Under stable and strictly positive diffusion, the bulk covariance Σ_{II} is positive-definite and invertible, so the subtraction is well-defined. However, in infinite dimensions, the stationary covariance operator is trace-class and compact, meaning its restriction $\Sigma_{\lambda,II}$ lacks a bounded global inverse. The appropriate operator-theoretic replacement is the shorted operator: for a non-negative block operator, the Schur complement is recovered as the quadratic form of the short to the target subspace, with the inverse term replaced by a variational supremum [15, 1]. To lift the Schur complement to infinite dimensions in the present singular-conditioning problem, we formulate the projection residual directly using covariance-energy forms.

Let $Q_0 \coloneqq \Sigma_{II}^{(0)}$ denote the fixed background bulk covariance under full decoupling ($\lambda = \infty$). We define the core Cameron–Martin space $H_{Q_0} \coloneqq \mathrm{Range}(Q_0^{1/2})$ and the core energy norm:

$$|u|_{Q_0} \coloneqq \|Q_0^{-1/2}u\| \quad \text{for all } u \in H_{Q_0}. \tag{18}$$

By Assumption 6, the off-diagonal drift coupling vector satisfies $a_{Ik} \in H_{Q_0}$, and the restricted drift generator $\mathcal{A}_{II}$ generates an exponentially stable semigroup on H_{Q_0}.

Under this assumption, the exact cross-covariance resolvent formula (derived in Section A) satisfies:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1}a_{Ik}. \tag{19}$$

This resolvent response yields the Cameron–Martin energy bound on the cross-covariance block:

$$|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}) \quad \text{as } \lambda \to \infty. \tag{20}$$

To define the infinite-dimensional shorted residual, we introduce the variational Schur subtraction R_λ:

$$R_\lambda \coloneqq \sup_{\langle u,\Sigma_{\lambda,II}u\rangle>0} \frac{|\langle u, \Sigma_{\lambda,Ik}\rangle|^2}{\langle u, \Sigma_{\lambda,II}u\rangle}. \tag{21}$$

Equivalently, if Σ_λ is viewed as a non-negative block operator on $\mathbb{C}\phi_k \oplus \mathcal{H}_I$, then (21) is the one-dimensional Anderson–Trapp shorted-operator subtraction, namely

$$\left\langle \phi_k, \mathcal{S}_{\mathbb{C}\phi_k}(\Sigma_\lambda)\phi_k \right\rangle = \Sigma_{\lambda,kk} - R_\lambda. \tag{22}$$

The restriction to vectors with $\langle u, \Sigma_{\lambda,II}u\rangle > 0$ simply removes null directions of the denominator; for a non-negative block operator, such directions do not change the supremum. Thus R_λ is not an ambient inverse computation but the energy removed by shorting to the target coordinate [1, 18].

Since the target noise channel and bulk noise channel are independent, the bulk state decomposes into independent background and target-driven components, meaning the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). Douglas range inclusion then guarantees that the variational subtraction is bounded:

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|_{Q_0}^2 = O(\lambda^{-6}). \tag{23}$$

3.2. **Shorted Schur Residual Scaling.** We now establish the main scaling theorem for the shorted Schur residual. The L^2 projection interpretation is secondary and follows from covariance geometry.

Theorem 9 (Hilbert Shorted Schur Invariant). *Under Assumptions 6 and 6, the L^2-projection residual satisfies:*

$$S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I} X_k\|_{L^2}^2 = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \tag{24}$$

where the remainder satisfies $0 \le R_\lambda = O(\lambda^{-6})$. In particular, the shorted Schur invariant is:

$$m_2^{\mathrm{Schur}} = \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda} = \frac{\varepsilon_0}{2}, \tag{25}$$

and the reciprocal projection-residual scaling satisfies:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}). \tag{26}$$

This result depends only on covariance and orthogonal projection; it holds for any linear stationary model satisfying these assumptions, without Gaussianity.

Proof. See Section A for the full derivation. □

3.3. **Convergence of Finite-Dimensional Truncations.** Let $\Pi_n : \mathcal{H} \to \mathcal{H}_n := \operatorname{span}\{\phi_1, \dots, \phi_n\}$ be the n-dimensional Galerkin projection. Write $\mathcal{A}_n^{(\lambda)} := \Pi_n \mathcal{A}^{(\lambda)} \Pi_n$, $\mathcal{D}_n^{(\lambda)} := \Pi_n \mathcal{D}^{(\lambda)} \Pi_n$, and let $\Sigma_{n,\lambda}$ solve the truncated Lyapunov equation $\mathcal{A}_n^{(\lambda)} \Sigma_{n,\lambda} + \Sigma_{n,\lambda} (\mathcal{A}_n^{(\lambda)})^* + \mathcal{D}_n^{(\lambda)} = 0$.

Proposition 10 (Galerkin Consistency of the Shorted Schur Invariant). *Under Assumption 6, let $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ be finite-rank orthogonal Galerkin projections preserving the target-bulk block decomposition:*

$$\Pi_n \phi_k = \phi_k, \qquad \Pi_n \mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \text{ strongly on } \mathcal{H}. \tag{27}$$

Assume the truncated regularized operator family satisfies the uniform exponential stability and trace-class conditions stated in Proposition 25 of Section A. Then the truncated covariance operators converge in trace norm:

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0 \quad \text{as } n \to \infty, \tag{28}$$

and the truncated Schur invariant is exact at every level:

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2} \quad \text{for all } n \ge k. \tag{29}$$

Proof. See Section A for the operator and spectral support. □

3.4. **The Triple Impossibility of Direct Inversion.** The variational shorting formula of Section 3.1 replaces the algebraic inverse Σ_{II}^{-1} by an energy projection. We now prove that this replacement is not merely elegant but structurally unavoidable: direct inversion fails at three independent and cumulative levels—algebraic, numerical, and asymptotic.

Theorem 11 (Triple Impossibility). *Let $Q_0 := \Sigma_{II}^{(0)}$ be the free-block covariance operator under full decoupling with eigenvalues $\{\sigma_j\}_{j=1}^\infty$. Let $N \ge 1$ and let $Q_0^{(N)}$ be the N-dimensional Galerkin truncation. Then:*

(i) ***Algebraic impossibility.*** *In infinite dimensions, Q_0 is either compact with 0 in its spectrum (GFF regime, $\sigma_j \sim j^{-2}$) or unbounded with no bounded global inverse (OU regime, $\sigma_j \sim j^2$ on the appropriate dual space). In either case, the classical Schur expression $\Sigma_{KK} - \Sigma_{KI}\Sigma_{II}^{-1}\Sigma_{IK}$ is not an operator formula in the norm topology of the ambient Hilbert space.*

(ii) ***Numerical impossibility.*** *For any finite-dimensional truncation of dimension N, the condition number of the truncated covariance matrix satisfies*

$$\kappa\big(Q_0^{(N)}\big) = \frac{\sigma_{\max}^{(N)}}{\sigma_{\min}^{(N)}} \sim N^2. \tag{30}$$

Consequently, for $N \gtrsim 10^3$, direct matrix inversion or Cholesky decomposition in IEEE-754 double precision suffers catastrophic cancellation (loss of $\sim 2\log_{10} N$ significant digits). At $N = 2048$, $\kappa \approx 4\times 10^6$ and the computed Schur complement bears no deterministic relation to the true infinite-dimensional limit. Extraction of the correct $O(\lambda^{-6})$ residual then requires quadruple-precision (256-bit) arithmetic.

(iii) ***Asymptotic impossibility.*** *Even with exact arithmetic, the finite-dimensional regularized Schur complement exhibits a* transient spectral imbalance *whose severity depends critically on the spectral regime:*

- ***GFF regime*** *($\mathcal{A}_0 = -L$, $\sigma_j \sim j^{-2}$): The resolvent $(\lambda I - \mathcal{A}_{II})^{-1}$ enters the asymptotic regime only when $\lambda \gg \sigma_{\min}^{-1} \sim N^2$. For $\lambda \ll N^2$, modes with $j \gtrsim \sqrt{\lambda}$ remain unpenalized, producing pre-asymptotic growth $\lambda^4 \operatorname{Tr}(\rho_E) = O(\lambda^2)$.*
- ***OU regime*** *($\mathcal{A}_0 = -L^{-1}$, $\sigma_j \sim j^2$): The resolvent $(\lambda I + L^{-1})^{-1}$ is equivalent to $L(L + \tilde{\lambda} I)^{-1}$ with $\tilde{\lambda} = \lambda^{-1}$. The critical mode satisfies $\tilde{\lambda}\sigma_{j^*} \sim 1$, i.e., $j^* \sim \tilde{\lambda}^{-1/2} = \sqrt{\lambda}$. Because $\tilde{\lambda} \to 0$ as $\lambda \to \infty$, all modes enter the asymptotic regime uniformly. The transient growth is eliminated entirely; convergence is immediate.*

In the GFF regime, no fixed finite-dimensional truncation can simultaneously approximate the infinite-dimensional operator and capture the large-λ limit.

The only consistent resolution in both regimes is the variational shorting operator (Section 3.1), which is well-defined without bounded inverses and numerically realizable through sparse precision-operator solves.

Proof. (i) follows from standard spectral theory: a compact operator on an infinite-dimensional Hilbert space has 0 in its spectrum, and an unbounded operator has no bounded global inverse.

For (ii), the eigenvalue ratio $\sigma_{\max}^{(N)}/\sigma_{\min}^{(N)}$ scales as N^2 in both the GFF regime ($\sigma_{\max} \sim 1$, $\sigma_{\min} \sim N^{-2}$) and the OU regime ($\sigma_{\max} \sim N^2$, $\sigma_{\min} \sim 1$).

In double precision ($\varepsilon_{\text{mach}} \approx 2 \times 10^{-16}$), the effective relative accuracy after inversion is $\varepsilon_{\text{mach}} \cdot \kappa \sim 10^{-9}$ at $N = 2048$, insufficient for extracting $O(\lambda^{-6})$ residuals.

For (iii), the GFF analysis follows from the resolvent diagonalization $(\lambda + \mu_j)^{-1}$ with $\mu_j \sim j^2$. The critical mode is $j^*(\lambda) \sim \pi^{-1}\sqrt{\lambda}$; modes with $j > j^*$ remain in the μ_j^{-1} regime and are unpenalized by λ, producing transient growth until $\lambda \gg N^2$.

For the OU regime, the algebraic identity

$$(\lambda I + L^{-1})^{-1} = L(\lambda L + I)^{-1} = \frac{1}{\lambda} L(L + \tilde{\lambda} I)^{-1}, \qquad \tilde{\lambda} = \lambda^{-1},$$

shows that the resolvent is realized by a *sparse* linear solve with the precision operator L at the *conjugate* regularization $\tilde{\lambda}$. Because $\tilde{\lambda} \to 0$, the system $(L + \tilde{\lambda} I)y = Lb$ is well-conditioned for all $\lambda \geq 1$, and all spectral modes converge uniformly. □

Remark 12 (On the necessity of extended precision). In the numerical verification reported in Section C, the quantity $\lambda^4 \operatorname{Tr}(\rho_{\lambda,E})$ at $\lambda \sim 10^5$ is of order 10^{-20}. Double-precision arithmetic ($\sim 10^{-16}$ relative accuracy) cannot resolve this residual against numerical noise. Combined with the condition-number amplification $\kappa \sim 10^6$ from (30), the effective signal-to-noise ratio is below unity. Extraction of the true $O(\lambda^{-4})$ rate therefore required quadruple-precision (256-bit) arithmetic—a computational necessity that confirms, rather than refutes, the numerical impossibility claim in Theorem 11(ii).

4. Fredholm Trace Anomalies and Gaussian Finite Parts

This section applies the inverse-free Schur construction to Gaussian Radon laws. Disintegration supplies the exact-evidence measure; the new algebraic content is the cancellation between the Fredholm determinant term and the trace anomaly generated by the shorted covariance defect.

4.1. **Radon Gaussian Application and Disintegration.**

Assumption 13 (Gaussian Application and Affine Mean Channel)**.** In addition to Assumptions 6 and 6, let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Gaussian Radon law on $\mathcal{H}$ with injective trace-class covariance. Let μ_{obs} be the canonical Gaussian disintegration member on the evidence hyperplane $X_k = x_k^*$. Suppose that a fixed deterministic forcing $b \in \mathcal{H}$ gives the stationary mean equation:

$$\mathcal{A}_{\text{cond}} m_\lambda + b - \lambda \Pi_k (m_\lambda - x_k^* \phi_k) = 0, \quad \Pi_k \mathcal{A}_{\text{cond}} = 0. \tag{31}$$

We define the target deterministic offset constant:

$$c_\lambda := \lambda(m_{\lambda,k} - x_k^*) = b_k, \tag{32}$$

where $m_{\lambda,k} := \langle m_\lambda, \phi_k \rangle$ and $b_k := \langle b, \phi_k \rangle$.

Under the Gaussian Radon law, the coordinate projection $\pi_k : \mathcal{H} \to \mathbb{R}$ is continuous, so the target variable X_k is a well-defined Borel random variable. The Polish structure of the Hilbert space guarantees the existence of a regular conditional probability distribution. By selecting the weakly continuous version of the conditional law at the single point $t = x_k^*$ (as detailed in Section B), we obtain the unique pointwise disintegration member μ_{obs}. Under this measure, the target coordinate satisfies the exact target zeros:

$$m_{\text{obs},k} - x_k^* = 0, \quad \text{Var}_{\text{obs}}(X_k) = 0 \quad \text{(exact)}. \tag{33}$$

4.2. **Measurable Cameron–Martin Energy Projection.** Let $C_\lambda := \Sigma_{\lambda,II}$ denote the Gaussian bulk covariance operator on $\mathcal{H}_I$. Because C_λ is trace-class and compact, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} := \text{Range}(C_\lambda^{1/2})$ and the corresponding energy norm.

The leakage cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm $\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} = O(\lambda^{-3})$. This range inclusion allows us to define the energy projection $\mu_{k|I}(X_I) := \mathbb{E}[X_k \mid X_I]$ as a measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{34}$$

which is the unique linear L^2 predictor. The estimates in Section B imply that, for all sufficiently large λ, the bulk marginals $\mu_{\text{obs},I}$ and $\mu_{\lambda,I}$ are equivalent and have finite bulk relative entropy:

$$D_{\text{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \text{as } \lambda \to \infty. \tag{35}$$

4.3. **Gaussian Fredholm Finite-Part Functional.** Although the joint KL divergence diverges under exact conditioning, the block-operator decomposition isolates a finite Fredholm part.

Definition 14 (Gaussian Fredholm Finite-Part Functional)**.** Whenever $S_{k,\lambda} > 0$ and $\mu_{\text{obs},I}$ is equivalent to $\mu_{\lambda,I}$, we define the Gaussian Fredholm finite-part functional of the conditioned law:

$$\begin{aligned} \mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) :=& D_{\text{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) \\ &+ \frac{1}{2} S_{k,\lambda}^{-1} \mathbb{E}_{\mu_{\text{obs}}} \left[\left(X_k - \mu_{k|I}(X_I) \right)^2 \right]. \end{aligned} \tag{36}$$

This is the finite part of the joint KL divergence after subtracting the Dirac normalization singularity; the remaining covariance contribution is controlled by Fredholm determinant and trace terms.

Weak scheme-independence. For an evidence level $t \in \mathbb{R}$, write μ_λ^t for the weakly continuous conditional law from Section B and set

$$\mathfrak{J}_\lambda(t) := \mathfrak{J}_\lambda(\mu_\lambda^t; \mu_\lambda).$$

Then, for any two evidence levels t_1, t_2 for which the finite part is defined,

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{(t_2 - m_{\lambda,k})^2 - (t_1 - m_{\lambda,k})^2}{2\Sigma_{\lambda,kk}}. \tag{37}$$

In the calibrated scaling $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$, if $b_i = \lambda(m_{\lambda,k} - t_i)$, this becomes

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{b_2^2 - b_1^2}{\varepsilon_0}.$$

Proof. Let $\Delta_t \coloneqq t - m_{\lambda,k}$ and

$$\rho_\lambda \coloneqq \frac{\|\Sigma_{\lambda,Ik}\|^2_{H_{C_\lambda}}}{\Sigma_{\lambda,kk}} = 1 - \frac{S_{k,\lambda}}{\Sigma_{\lambda,kk}}.$$

The Gaussian entropy computation in Section B gives

$$D_{\mathrm{KL}}(\mu^t_{\lambda,I} \parallel \mu_{\lambda,I}) = \frac{1}{2}\frac{\Delta_t^2}{\Sigma_{\lambda,kk}}\rho_\lambda - \frac{1}{2}\log(1-\rho_\lambda) - \frac{1}{2}\rho_\lambda.$$

The conditional-residual term in (36) is

$$\frac{1}{2S_{k,\lambda}}\mathbb{E}_{\mu^t_\lambda}\left[(X_k - \mu_{k|I}(X_I))^2\right] = \frac{1}{2}\rho_\lambda + \frac{1}{2}\frac{\Delta_t^2}{\Sigma_{\lambda,kk}}(1-\rho_\lambda).$$

Adding these two identities cancels the $\rho_\lambda/2$ terms and yields

$$\mathfrak{J}_\lambda(t) = \frac{(t - m_{\lambda,k})^2}{2\Sigma_{\lambda,kk}} - \frac{1}{2}\log(1-\rho_\lambda).$$

The logarithmic term depends on the reference covariance and the conditioning scale, but not on the evidence value t. Hence it cancels in the difference between t_2 and t_1, proving (37). The same argument shows weak independence from the mollifying kernel: replacing the Gaussian Dirac mollifier in Theorem 30 by any centered approximate identity with finite second moment changes the extracted finite part only by an additive term depending on the kernel, λ, and $S_{k,\lambda}$, not on t. Therefore all evidence differences $\Delta\mathfrak{J}_\lambda$ are kernel-independent. □

4.4. **Gaussian Main Theorem.** We now establish the main Fredholm regularization and cancellation theorems for the finite-part functional.

Theorem 15 (Gaussian Fredholm Finite-Part Regularization)**.** *Under Assumption 13, there exists $\lambda_0 > 0$ such that, for $\lambda \geq \lambda_0$, the Gaussian Fredholm finite-part functional under the conditioned law μ_{obs} is well-defined and satisfies:*

$$\mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}). \tag{38}$$

In particular, the functional remains bounded at $O(1)$ and converges to:

$$\lim_{\lambda\to\infty}\mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda) = \frac{b_k^2}{\varepsilon_0}. \tag{39}$$

Proof. See Section B for the full derivation. □

Corollary 16 (Exact Decoupled-Centered Cancellation)**.** *Under Assumption 13, if the target coordinate is decoupled from the bulk ($a_{Ik} = 0$) and the evidence is centered at the stationary mean ($x_k^* = m_{\lambda,k}$), then:*

$$\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = 0 \quad \textit{for every } \lambda > 0. \tag{40}$$

Proof. The decoupling implies $\Sigma_{\lambda,Ik} = 0$, so $D_{\text{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) = 0$. The centering makes the conditional residual zero under the conditioning, so both terms in (36) vanish. □

5. Elliptic Gaussian Field Example

We now record a standard elliptic Gaussian field example to show that the abstract Hilbert-space assumptions are nonempty and natural for covariance operators arising from stochastic evolution equations and elliptic Gaussian priors [8, 17, 21].

5.1. Elliptic Gaussian field on $[0, 1]$.

Let

$$L = -\frac{d^2}{dx^2} + \kappa^2$$

on $L^2(0, 1)$ with Dirichlet boundary conditions and $\kappa > 0$. Then $\mathcal{A}_0 = -L$ with domain $H^2(0, 1) \cap H_0^1(0, 1)$ is dissipative, L is positive self-adjoint and sectorial, and e^{-tL} is an exponentially stable analytic contraction semigroup. The inverse covariance

$$Q_0 = L^{-1}$$

is compact, positive, self-adjoint, and trace class. With eigenfunctions $e_j(x) = \sqrt{2}\sin(\pi j x)$, its eigenvalues are

$$q_j = (\pi^2 j^2 + \kappa^2)^{-1}.$$

Thus Q_0 is a typical infinite-dimensional covariance operator: it is compact and trace class, has no bounded inverse on the ambient space, and carries its inverse only as a Cameron–Martin energy form [2, 19].

5.2. Cameron–Martin range and covariance energy.

The Cameron–Martin space associated with Q_0 is

$$H_{Q_0} = \text{Range}(Q_0^{1/2}) = H_0^1(0, 1),$$

with equivalent energy norm

$$\|h\|_{Q_0}^2 = \langle h, Lh \rangle_{L^2} = \int_0^1 \left(|h'(x)|^2 + \kappa^2 |h(x)|^2\right) dx.$$

This identifies the variational Schur projection in a Sobolev-energy form, as in the standard covariance-energy representation of Gaussian Hilbert priors [2, 21]. The covariance inverse does not act as a bounded operator on $L^2(0, 1)$; it acts only through the energy form on its Cameron–Martin domain.

5.3. **Spectral sensor conditioning.** Fix a target eigenmode e_k and define the observed coordinate

$$X_k = \langle X, e_k \rangle_{L^2}.$$

The unperturbed target variance is

$$\Sigma_{0,kk} = q_k = (\pi^2 k^2 + \kappa^2)^{-1}.$$

A regularized restoring drift on this coordinate produces the exact normalization

$$\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad R_\lambda = O(\lambda^{-6}),$$

under the block assumptions of Section 3. Off-diagonal bounded perturbations of the drift operator generate the cross-covariance vector entering the variational Schur residual. The resulting subtraction is evaluated by the Cameron–Martin energy form above, not by a global inverse on $L^2(0,1)$.

This example shows that the abstract theorem applies to a genuine Gaussian field. The residual is controlled by a Sobolev energy on $H_0^1(0,1)$ while the ambient covariance remains compact.

6. Finite-Rank Shorting and Exact Fredholm Trace Cancellation

We have resolved the rank-one block-operator obstruction: a compact free covariance block need not be inverted, because the Schur subtraction is the Anderson–Trapp shorted residual on the Cameron–Martin energy range. Finite-rank targets require the same idea in operator-valued form, plus an exact matrix/Fredholm identity for the Gaussian finite part. Here Galerkin approximation remains purely variational: truncation restricts the Anderson–Trapp supremum to nested finite-dimensional test spaces. The limitation is not finite rank, but infinite-rank observation, where Fredholm determinants, trace-class perturbation hypotheses, and operator-valued finite parts need separate treatment.

6.1. **Matrix algebraic extension.** For an m-dimensional observed subspace $E = \operatorname{span}\{\phi_1, \dots, \phi_m\}$, the scalar target block $\Sigma_{\lambda,kk}$ is replaced by the covariance matrix $\Sigma_{\lambda,EE} \in \mathbb{R}^{m\times m}$. The scalar calibration becomes a matrix Lyapunov balancing problem.

Theorem 17 (Matrix Lyapunov Calibration)**.** *Let E be an m-dimensional target subspace. Suppose the regularized drift and diffusion on E are given by $A_{\lambda,E} = -\lambda\Lambda_E$ and $D_{\lambda,E} = \frac{1}{\lambda}\Delta_E$, where $\Lambda_E, \Delta_E \in \mathbb{R}^{m\times m}$ are strictly positive definite matrices. Let $\Sigma_{\lambda,EE}$ be the unique positive definite solution to the matrix Lyapunov equation:*

$$A_{\lambda,E}\Sigma_{\lambda,EE} + \Sigma_{\lambda,EE}A^*_{\lambda,E} + D_{\lambda,E} = 0. \tag{41}$$

Then $\Sigma_{\lambda,EE}$ scales exactly as λ^{-2}. Specifically, the normalized matrix $\widetilde{\Sigma}_{EE} := \lambda^2\Sigma_{\lambda,EE}$ is independent of λ and is the unique positive definite solution to *the scale-free Lyapunov equation:*

$$\Lambda_E \widetilde{\Sigma}_{EE} + \widetilde{\Sigma}_{EE} \Lambda_E^* = \Delta_E. \tag{42}$$

Proof. Substituting $A_{\lambda,E} = -\lambda \Lambda_E$ and $D_{\lambda,E} = \frac{1}{\lambda}\Delta_E$ into (41) yields:

$$-\lambda \Lambda_E \Sigma_{\lambda,EE} - \lambda \Sigma_{\lambda,EE} \Lambda_E^* + \frac{1}{\lambda}\Delta_E = 0.$$

Multiplying the entire equation by λ gives:

$$\Lambda_E(\lambda^2 \Sigma_{\lambda,EE}) + (\lambda^2 \Sigma_{\lambda,EE})\Lambda_E^* = \Delta_E.$$

Defining $\widetilde{\Sigma}_{EE} \coloneqq \lambda^2 \Sigma_{\lambda,EE}$, we recover (42). Since Λ_E is positive definite, its spectrum lies in the open right half-plane, making $-\Lambda_E$ Hurwitz. Because Δ_E is positive definite, standard continuous Lyapunov theory guarantees that (42) admits a unique positive definite solution $\widetilde{\Sigma}_{EE}$. Therefore, the exact scaling $\Sigma_{\lambda,EE} = \lambda^{-2}\widetilde{\Sigma}_{EE}$ holds for all $\lambda > 0$, properly generalizing the scalar calibration condition $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$. □

6.2. **Inverse-Free Residuals and Loewner Monotonicity.** For an m-dimensional target space E, the Schur residual is a positive-semidefinite operator on E. Since the free-block covariance $C_\lambda \coloneqq \Sigma_{\lambda,II}$ acts as a compact, trace-class operator on the infinite-dimensional space, it admits no bounded global inverse. Thus the formal expression

$$\Sigma_{\lambda,EI} C_\lambda^{-1} \Sigma_{\lambda,IE}$$

is not used as a definition. The residual is defined instead by the Anderson–Trapp variational shorting formula.

Theorem 18 (Inverse-free operator residual and Galerkin Loewner monotonicity)**.** *Let $\mathcal{H} = E \oplus \mathcal{H}_I$, where E is finite dimensional. Let $C_\lambda = \Sigma_{\lambda,II}$ be a non-negative trace-class covariance operator on $\mathcal{H}_I$, and set*

$$B_\lambda \coloneqq \Sigma_{\lambda,IE} : E \to \mathcal{H}_I.$$

Assume the Cameron–Martin range condition

$$B_\lambda E \subset H_{C_\lambda} \coloneqq \mathrm{Ran}(C_\lambda^{1/2}).$$

Define the shorted residual $R_{\lambda,E}$ on E by

$$\langle v, R_{\lambda,E} v\rangle_E = \sup_{\substack{u \in \mathcal{H}_I \\ \langle u, C_\lambda u\rangle > 0}} \frac{|\langle u, B_\lambda v\rangle_{\mathcal{H}_I}|^2}{\langle u, C_\lambda u\rangle_{\mathcal{H}_I}}, \qquad v \in E. \tag{43}$$

Then $R_{\lambda,E}$ is a well-defined bounded positive-semidefinite operator on E. If

$$\|B_\lambda v\|_{H_{C_\lambda}} \le M\lambda^{-3}\|v\|_E, \qquad v \in E,$$

then

$$\|R_{\lambda,E}\|_{\mathcal{L}(E)} = O(\lambda^{-6}).$$

Let $(\mathcal{H}_n)_{n\geq 1}$ be an increasing sequence of finite-dimensional subspaces of $\mathcal{H}_I$ with dense union. Define the Galerkin residuals by the restricted variational problems

$$\langle v, R^{(n)}_{\lambda,E} v\rangle_E := \sup_{\substack{u\in\mathcal{H}_n \\ \langle u, C_\lambda u\rangle>0}} \frac{|\langle u, B_\lambda v\rangle_{\mathcal{H}_I}|^2}{\langle u, C_\lambda u\rangle_{\mathcal{H}_I}}, \qquad v\in E. \tag{44}$$

Then

$$0 \preceq R^{(n)}_{\lambda,E} \preceq R^{(n+1)}_{\lambda,E} \preceq R_{\lambda,E}$$

in the Loewner order, and

$$R^{(n)}_{\lambda,E} \longrightarrow R_{\lambda,E}$$

in operator norm on E.

Proof. Fix $v \in E$ and write $y_v = B_\lambda v$. By the range assumption, $y_v \in H_{C_\lambda}$. Thus the Cameron–Martin norm $\|y_v\|_{H_{C_\lambda}}$ is finite. Equivalently, the quadratic-form Cauchy–Schwarz inequality gives

$$|\langle u, y_v\rangle_{\mathcal{H}_I}|^2 \leq \|y_v\|^2_{H_{C_\lambda}} \langle u, C_\lambda u\rangle_{\mathcal{H}_I}, \qquad u\in\mathcal{H}_I.$$

Hence the supremum in (43) is finite and equals

$$\|y_v\|^2_{H_{C_\lambda}}.$$

Therefore $v \mapsto \langle v, R_{\lambda,E} v\rangle_E$ defines a finite non-negative quadratic form on the finite-dimensional space E, and is thus represented by a unique positive-semidefinite operator $R_{\lambda,E}$. The same argument applies to each restricted supremum and defines $R^{(n)}_{\lambda,E}$.

Since

$$\mathcal{H}_n \subset \mathcal{H}_{n+1} \subset \mathcal{H}_I,$$

the admissible class in the restricted variational problem increases with n. Hence, for every $v \in E$,

$$\langle v, R^{(n)}_{\lambda,E} v\rangle_E \leq \langle v, R^{(n+1)}_{\lambda,E} v\rangle_E \leq \langle v, R_{\lambda,E} v\rangle_E.$$

This is precisely the Loewner monotonicity.

It remains to identify the limit. Let

$$q_n(v) = \langle v, R^{(n)}_{\lambda,E} v\rangle_E, \qquad q(v) = \langle v, R_{\lambda,E} v\rangle_E.$$

The sequence $q_n(v)$ is increasing and bounded above by $q(v)$. Conversely, fix any admissible $u \in \mathcal{H}_I$ with $\langle u, C_\lambda u\rangle > 0$. Since $\bigcup_n \mathcal{H}_n$ is dense, there exist $u_n \in \mathcal{H}_n$ with $u_n \to u$ in $\mathcal{H}_I$. Because C_λ is bounded and y_v is fixed,

$$\langle u_n, C_\lambda u_n\rangle \to \langle u, C_\lambda u\rangle, \qquad \langle u_n, y_v\rangle \to \langle u, y_v\rangle.$$

Thus every value of the full Rayleigh quotient can be approximated by values taken over Galerkin subspaces. Taking suprema gives

$$\lim_{n\to\infty} q_n(v) = q(v), \qquad v\in E.$$

Since E is finite dimensional, pointwise convergence of the quadratic forms implies convergence of their representing matrices in operator norm.

Finally, the assumed Cameron–Martin energy estimate gives

$$\langle v, R_{\lambda,E} v\rangle_E = \|B_\lambda v\|^2_{H_{C_\lambda}} \le M^2\lambda^{-6}\|v\|^2_E,$$

and hence

$$\|R_{\lambda,E}\|_{\mathcal{L}(E)} \le M^2\lambda^{-6}.$$

□

Remark 19 (Why strict Loewner growth is not expected)*.* The monotonicity in Theorem 18 is non-decreasing Loewner monotonicity. Strict Loewner growth is not structurally guaranteed.

Indeed, suppose that the Galerkin space is enlarged by a finite-dimensional increment

$$\Delta\mathcal{H}_n = \mathcal{H}_{n+1} \ominus \mathcal{H}_n$$

of dimension k. In the Cameron–Martin energy geometry, the additional Rayleigh–Ritz information available at the next step has rank at most k. Consequently

$$\operatorname{rank}\big(R^{(n+1)}_{\lambda,E} - R^{(n)}_{\lambda,E}\big) \le k.$$

If the Galerkin scheme adds only one new direction at each step, then the increment is at most rank one. For a target space of dimension $m > 1$, a rank-one positive-semidefinite increment cannot be positive definite on E. Therefore

$$R^{(n+1)}_{\lambda,E} - R^{(n)}_{\lambda,E} \succ 0$$

is impossible for one-dimensional Galerkin updates when $m > 1$.

Strict positivity of an increment requires both enough new Galerkin directions, $k \ge m$, and full-rank visibility of the unresolved cross-covariance defect across all target directions.

6.3. **General observation operators and misalignment.** The scalar proof in this paper uses a block decomposition aligned with a single observed coordinate. However, typical finite-rank observation operators $L : \mathcal{H} \to \mathbb{R}^m$ select a subspace E that does not commute with the free prior covariance, meaning the target-free split is not an eigenbasis split. This geometric misalignment generates cross-coupling that requires a precise resolvent analysis.

Theorem 20 (Misaligned Covariance Coupling)**.** *Let P_E be an orthogonal projection onto an m-dimensional observed subspace E, and let $\mathcal{H}_I = E^\perp$. Let the regularized operator family be decomposed such that the target block matches Theorem 17 with $A_{\lambda,E} = -\lambda\Lambda_E$, the cross-noise is zero ($D_{\lambda,IE} = 0$), and the off-diagonal drift coupling is denoted by $A_{\lambda,IE}$. Assume that, for some $\lambda_0 > 0$,*

$$\sup_{\lambda\ge\lambda_0} \|A_{\lambda,IE}\|_{\mathcal{L}(E,H_{Q_0})} < \infty.$$

Assume also that the restricted drift A_{II} generates an exponentially stable semigroup on H_{Q_0} and that there exist $K, \gamma > 0$ such that

$$\|e^{-t\Lambda_E^*}\|_{\mathcal{L}(E)} \leq Ke^{-\gamma t}, \qquad t \geq 0.$$

Then the cross-covariance satisfies the exact identity:

$$\Sigma_{\lambda,IE} = \int_0^\infty e^{tA_{II}} A_{\lambda,IE} \Sigma_{\lambda,EE} e^{tA_{\lambda,E}^*} dt. \tag{45}$$

Furthermore, the misaligned cross-covariance energy scales as $\|\Sigma_{\lambda,IE} v\|_{H_{Q_0}} = O(\lambda^{-3})$ for any $v \in E$.

Proof. The off-diagonal block of the global Lyapunov equation $\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0$ yields the Sylvester equation:

$$A_{II}\Sigma_{\lambda,IE} + \Sigma_{\lambda,IE}A_{\lambda,E}^* + A_{\lambda,IE}\Sigma_{\lambda,EE} = 0.$$

Since A_{II} generates an exponentially stable semigroup $e^{tA_{II}}$ and the assumed bound controls $e^{tA_{\lambda,E}^*} = e^{-\lambda t\Lambda_E^*}$, the unique solution is given by the integral representation (45).

To bound the Cameron–Martin norm, fix $v \in E$. We have $e^{tA_{\lambda,E}^*}v = e^{-\lambda t\Lambda_E^*}v$. By Theorem 17, $\Sigma_{\lambda,EE} = O(\lambda^{-2})$. By the uniform coupling assumption, choose C such that $\|A_{\lambda,IE}w\|_{H_{Q_0}} \leq C\|w\|_E$ for all $w \in E$ and all $\lambda \geq \lambda_0$. Using the uniform exponential stability of $e^{tA_{II}}$ on H_{Q_0}, namely $\|e^{tA_{II}}\|_{\mathcal{L}(H_{Q_0})} \leq Me^{-\omega t}$ for some $M \geq 1, \omega > 0$, we estimate:

$$\begin{aligned}
\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} &\leq \int_0^\infty \|e^{tA_{II}}\|_{\mathcal{L}(H_{Q_0})} \|A_{\lambda,IE}\Sigma_{\lambda,EE}e^{-\lambda t\Lambda_E^*}v\|_{H_{Q_0}} dt \\
&\leq \int_0^\infty Me^{-\omega t} C\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)} \|e^{-\lambda t\Lambda_E^*}\|_{\mathcal{L}(E)} \|v\|_E dt.
\end{aligned}$$

Using $\|e^{-\lambda t\Lambda_E^*}\|_{\mathcal{L}(E)} \leq Ke^{-\lambda\gamma t}$, we obtain:

$$\begin{aligned}
\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} &\leq MCK\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)}\|v\|_E \int_0^\infty e^{-(\omega+\lambda\gamma)t} dt \\
&= MCK\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)}\|v\|_E \frac{1}{\omega + \lambda\gamma}.
\end{aligned}$$

Since $\|\Sigma_{\lambda,EE}\|_{\mathcal{L}(E)} = O(\lambda^{-2})$ and $(\omega + \lambda\gamma)^{-1} = O(\lambda^{-1})$, we conclude that $\|\Sigma_{\lambda,IE}v\|_{H_{Q_0}} = O(\lambda^{-3})$. Combined with Theorem 18, this verifies that the shorted-operator residual maintains the $O(\lambda^{-6})$ scaling even in the presence of observation misalignment. $\square$

6.4. **Multidimensional Fredholm Trace-Anomaly Cancellation.** With the finite-rank covariance bounds in Theorems 17, 18 and 20 in place, the rank-one finite-part functional from Section 4 becomes a matrix problem. The determinant contribution is no longer a scalar logarithm expansion; it is a Fredholm determinant reduced to an $m \times m$ matrix by the Weinstein–Aronszajn identity. The trace anomaly then cancels exactly against the matrix-valued shorted residual term.

The scalar renormalized functional can be fully generalized to an m-dimensional observation subspace E. Let the exact evidence be $X_E = x_E^* \in \mathbb{R}^m$. The true Schur residual matrix is $S_{\lambda,E} := \Sigma_{\lambda,EE} - R_{\lambda,E} \in \mathbb{R}^{m\times m}$, where $R_{\lambda,E}$ is the shorted-operator residual from Theorem 18. Since $R_{\lambda,E}^{(n)} \nearrow R_{\lambda,E}$ in the Loewner order, the Galerkin residual matrices

$$S_{\lambda,E}^{(n)} = \Sigma_{\lambda,EE} - R_{\lambda,E}^{(n)}$$

decrease in the Loewner order to

$$S_{\lambda,E} = \Sigma_{\lambda,EE} - R_{\lambda,E}.$$

They are not claimed to decrease strictly. Positivity of the limiting matrix is guaranteed by the condition

$$\rho_{\lambda,E} = \Sigma_{\lambda,EE}^{-1/2} R_{\lambda,E} \Sigma_{\lambda,EE}^{-1/2} < I.$$

Definition 21 (Multidimensional Fredholm Finite-Part Functional)**.** Whenever $S_{\lambda,E} \succ 0$ and $\mu_{\mathrm{obs},I}$ is equivalent to $\mu_{\lambda,I}$, the multidimensional Gaussian Fredholm finite-part functional is defined by evaluating the conditional residual in the metric of the shorted Schur matrix:

(46)
$$\begin{aligned}\mathfrak{J}_{\lambda,E}(\mu_{\mathrm{obs}};\mu_\lambda) :=& D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \,\|\, \mu_{\lambda,I}) \\ &+ \frac{1}{2}\mathbb{E}_{\mu_{\mathrm{obs}}}\left[\left(X_E - \mu_{E|I}(X_I)\right)^T S_{\lambda,E}^{-1}\left(X_E - \mu_{E|I}(X_I)\right)\right],\end{aligned}$$

where $\mu_{E|I}(X_I) := m_{\lambda,E} + W_{\Sigma_{\lambda,IE}}(X_I - m_{\lambda,I})$ is the measurable Cameron–Martin energy projection on $\mathbb{R}^m$, with the Wiener integral evaluated componentwise.

To establish the multidimensional limit, we define the dimensionless coupling matrix $\rho_{\lambda,E} := \Sigma_{\lambda,EE}^{-1/2} R_{\lambda,E} \Sigma_{\lambda,EE}^{-1/2} \in \mathbb{R}^{m\times m}$. By the energy bound in Theorem 18, $\|\rho_{\lambda,E}\| = O(\lambda^{-4})$.

Theorem 22 (Exact Fredholm Trace and Matrix Determinant Cancellation)**.** *For any evidence level $t \in \mathbb{R}^m$, let μ_λ^t be the weakly continuous conditional law on $X_E = t$. The infinite-dimensional KL divergence and the matrix-valued shorted residual expectation undergo exact cancellation of all cross-coupling traces and quadratic forms, yielding:*

$$\mathfrak{J}_{\lambda,E}(\mu_\lambda^t;\mu_\lambda) = \frac{1}{2}(t - m_{\lambda,E})^T \Sigma_{\lambda,EE}^{-1}(t - m_{\lambda,E}) - \frac{1}{2}\log\det(I - \rho_{\lambda,E}). \tag{47}$$

Proof. Let $C_\lambda := \Sigma_{\lambda,II}$. Under μ_λ^t, the free coordinates follow $\mathcal{N}(\mu_{I|E}(t), \Sigma_{I|E})$. By the Feldman–Hájek formula for trace-class perturbations, evaluating the KL divergence relies on the operator $K := C_\lambda^{-1/2}\Sigma_{\lambda,IE}\Sigma_{\lambda,EE}^{-1/2}$. By the Weinstein–Aronszajn identity, the non-zero spectrum of the infinite-dimensional operator KK^* matches that of the $m \times m$ matrix $K^*K = \rho_{\lambda,E}$.

This reduces the infinite-dimensional Fredholm determinant and trace exactly to the matrix manifold:

$$D_{\mathrm{KL}}(\mu^t_{\lambda,I} \parallel \mu_{\lambda,I}) = \frac{1}{2}\Big[(t-m_{\lambda,E})^T\Sigma^{-1}_{\lambda,EE}R_{\lambda,E}\Sigma^{-1}_{\lambda,EE}(t-m_{\lambda,E}) \\ - \log\det(I-\rho_{\lambda,E}) - \mathrm{Tr}(\rho_{\lambda,E})\Big].$$

For the residual expectation term, let $Z := t - \mu_{E|I}(X_I)$. Using block matrix identities, the conditional mean is $\mathbb{E}_{\mu^t_\lambda}[Z] = S_{\lambda,E}\Sigma^{-1}_{\lambda,EE}(t-m_{\lambda,E})$ and the conditional variance is $\mathrm{Var}_{\mu^t_\lambda}(Z) = R_{\lambda,E}\Sigma^{-1}_{\lambda,EE}S_{\lambda,E}$. Expanding the quadratic form $\frac{1}{2}\mathbb{E}[Z^TS^{-1}_{\lambda,E}Z]$ gives:

$$\frac{1}{2}\Big[(t-m_{\lambda,E})^T\Sigma^{-1}_{\lambda,EE}(t-m_{\lambda,E}) \\ - (t-m_{\lambda,E})^T\Sigma^{-1}_{\lambda,EE}R_{\lambda,E}\Sigma^{-1}_{\lambda,EE}(t-m_{\lambda,E}) + \mathrm{Tr}(\rho_{\lambda,E})\Big].$$

Summing the KL divergence and the quadratic residual form exactly cancels both the $\pm\frac{1}{2}\mathrm{Tr}(\rho_{\lambda,E})$ trace anomalies and the cross-covariance residual matrices, isolating (47). $\square$

Theorem 23 (Multidimensional Fredholm Finite-Part Regularization)**.** *Assume the m-dimensional regularized drift on E is centered at the observed value:*

$$A_{\lambda,E}(m_{\lambda,E} - x^*_E) + b_E = 0, \qquad A_{\lambda,E} = -\lambda\Lambda_E.$$

Defining the normalized displacement $\delta_E := \Lambda_E^{-1}b_E \in \mathbb{R}^m$, the mean deterministic offset satisfies

$$m_{\lambda,E} - x^*_E = \lambda^{-1}\Lambda_E^{-1}b_E = \lambda^{-1}\delta_E.$$

As $\lambda \to \infty$, the multidimensional Fredholm finite-part functional converges to a well-defined bounded limit governed by the scale-free Lyapunov solution $\widetilde{\Sigma}_{EE}$:

$$\lim_{\lambda\to\infty} \mathfrak{J}_{\lambda,E}(\mu_{\mathrm{obs}};\mu_\lambda) = \frac{1}{2}\delta_E^T\widetilde{\Sigma}^{-1}_{EE}\delta_E. \tag{48}$$

The approximation residual satisfies $\mathfrak{J}_{\lambda,E} = \frac{1}{2}\delta_E^T\widetilde{\Sigma}^{-1}_{EE}\delta_E + O(\lambda^{-4})$.

Proof. Substitute $t = x^*_E$ and $t - m_{\lambda,E} = -\lambda^{-1}\delta_E$ into Theorem 22. Under the canonical scaling,

$$\Sigma_{\lambda,EE} = \lambda^{-2}\widetilde{\Sigma}_{EE}, \qquad t - m_{\lambda,E} = -\lambda^{-1}\delta_E, \qquad \|\rho_{\lambda,E}\| = O(\lambda^{-4}).$$

Therefore the quadratic term yields exactly

$$\frac{1}{2}\delta_E^T\widetilde{\Sigma}^{-1}_{EE}\delta_E,$$

and the determinant term satisfies $\log\det(I-\rho_{\lambda,E}) = O(\lambda^{-4})$. Hence

$$\mathfrak{J}_{\lambda,E} = \frac{1}{2}\delta_E^T\widetilde{\Sigma}_{EE}^{-1}\delta_E + O(\lambda^{-4}).$$

$\square$

Remark 24 (Algebraic Mechanism of Uniform Positivity). The $O(\lambda^{-4})$ decay of the normalized coupling matrix $\|\rho_{\lambda,E}\|$ is not merely an asymptotic bound; it provides the essential algebraic mechanism ensuring the uniform positivity of the Fredholm finite part. By extracting the leading-order expansion of the cross-covariance resolvent under the canonical scaling $D_{\lambda,E} = \Delta_E/\lambda$, the trace of the coupling matrix converges exactly as:

$$\lim_{\lambda\to\infty} \lambda^4 \operatorname{Tr}(\rho_{\lambda,E}) = \frac{\omega}{d_{\mathrm{bg}}}\sum_{j=1}^{m}\frac{c_j^2\Delta_j}{\Lambda_j^3}. \tag{49}$$

This extreme asymptotic decoupling directly resolves the algebraic obstruction of non-strict Loewner growth noted in Theorem 19. The Anderson–Trapp formulation and its Galerkin approximations are employed purely to establish the existence of the limiting shorted operator on the Hilbert space via the Monotone Sequence Theorem. For any nested subspace sequence, the finite-dimensional operator residuals satisfy $0 \preceq R_{\lambda,E}^{(n)} \preceq R_{\lambda,E}$. Normalizing by the target covariance block $\Sigma_{\lambda,EE} = O(\lambda^{-2})$ yields the spectral bound:

$$I \succeq \Sigma_{\lambda,EE}^{-1/2} S_{\lambda,E}^{(n)} \Sigma_{\lambda,EE}^{-1/2} \succeq I - \rho_{\lambda,E}.$$

Because the limiting normalized coupling strictly vanishes as $\operatorname{Tr}(\rho_{\lambda,E}) = O(\lambda^{-4})$, the spectrum of every intermediate operator $S_{\lambda,E}^{(n)}$ is uniformly bounded from below by $1 - O(\lambda^{-4}) > 0$. The sequence of conditional covariance operators is therefore uniformly positive definite. This guarantees that the Gaussian conditioning remains well-posed along the entire approximation path, demonstrating that shorted operators provide the necessary and sufficient algebraic mechanism to define the Fredholm finite part at singular evidence, without requiring strict Loewner monotonicity. The exact $O(\lambda^{-4})$ scaling of the strict coupling trace is verified numerically to arbitrary precision in Section C. The necessity of extended-precision arithmetic is not a computational luxury but a direct consequence of the condition-number catastrophe (Theorem 11(ii)): at $N = 2048$, $\kappa \sim 4 \times 10^6$ destroys double-precision accuracy, and the transient spectral imbalance forces λ into a regime where $\lambda^4 \operatorname{Tr}(\rho_{\lambda,E})$ is below the machine-epsilon floor. Only quadruple-precision (256-bit) arithmetic can resolve the true $O(\lambda^{-4})$ decay from numerical noise.

6.5. **Non-Gaussian and time-dependent extensions.** The result is deliberately limited to finite-rank covariance block algebra and its Gaussian Radon application. It does not assert a general non-Gaussian conditioning theory. For non-Gaussian stationary processes, the projection residual

$S_{\lambda,E} = \Sigma_{\lambda,EE} - R_{\lambda,E}$ still has a second-moment interpretation, but Radon disintegration, finite-part functionals, and measure-class behavior require separate hypotheses [3, 14]. Time-dependent relaxation after conditioning, or transport distances between post-conditioning laws, belongs to a different problem. The contribution here is the operator-theoretic removal of the compact-block Schur obstruction.

7. Conclusion

The paper resolves the Schur-complement obstruction created by compact trace-class free covariance blocks. Since such blocks have no bounded ambient inverse, the residual is defined by Anderson–Trapp shorting on the Cameron–Martin range. Galerkin convergence follows from nested variational suprema, not algebraic inversion. The Gaussian conditioning statements then follow as consequences: disintegration supplies the exact-evidence law, and Fredholm determinant/trace identities cancel the finite-rank anomalies. The paper does not treat infinite-rank observation operators or non-Gaussian conditioning theory.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used Google DeepMind's Antigravity assistant in order to merge the LaTeX manuscripts, clean up cross-referencing tags, format mathematical notation, and edit the draft for style consistency. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Appendix A. Exact Block and Variational Shorting Analysis

This appendix derives the linear invariants from the block Lyapunov equation and proves convergence of the variational Galerkin shorting scheme. We do not use Laurent series or bounded ambient covariance inverses.

A.1. **Exact Target Covariance Block.** Write $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ with $\mathcal{H}_k = \mathbb{R}\phi_k$. The incoming target-coordinate block decoupling and the calibrated noise scale give the block operators:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & D_{II} \end{pmatrix}. \tag{50}$$

The block $\Sigma_{\lambda,Ik} \in \mathcal{H}_I$ denotes the vector representation of the target-to-bulk cross-covariance. The kk block of the Lyapunov equation is the scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0 \quad \Longrightarrow \quad \Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2}. \tag{51}$$

This equality holds exactly for all $\lambda > 0$.

A.2. **Exact Cross-Covariance Resolvent Formula.** The Ik block of the Lyapunov equation reduces to:

$$(\mathcal{A}_{II} - \lambda I)\Sigma_{\lambda,Ik} + a_{Ik}\Sigma_{\lambda,kk} = 0. \tag{52}$$

This yields the exact cross-covariance resolvent formula:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}. \tag{53}$$

A.3. **Cameron–Martin Energy Bound.** Let $Q_0 := \Sigma_{II}^{(0)}$ be the background bulk covariance. The Cameron–Martin space $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ has energy norm $|u|_{Q_0} := \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. Under Assumption 6, the restricted resolvent satisfies the energy bound:

$$|(\lambda I - \mathcal{A}_{II})^{-1}u|_{Q_0} \le \frac{M}{\lambda+\omega}|u|_{Q_0} \tag{54}$$

for all $u \in H_{Q_0}$. It follows from (53) that the cross-covariance has finite energy norm and decays as:

$$|\Sigma_{\lambda,Ik}|_{Q_0} \le \frac{\varepsilon_0 M}{2\lambda^2(\lambda+\omega)}|a_{Ik}|_{Q_0} = O(\lambda^{-3}). \tag{55}$$

A.4. **Variational Shorted Residual Proof.** We define the Anderson–Trapp variational shorted subtraction R_λ:

$$R_\lambda = \sup_{\langle u, \Sigma_{\lambda,II}u\rangle > 0} \frac{1}{\langle u, \Sigma_{\lambda,II}u\rangle}|\langle u, \Sigma_{\lambda,Ik}\rangle|^2. \tag{56}$$

Because the target and bulk noise channels are independent, the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). We then have:

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|_{Q_0}^2 \le \frac{\varepsilon_0^2 M^2 |a_{Ik}|_{Q_0}^2}{4\lambda^4(\lambda+\omega)^2} = O(\lambda^{-6}). \tag{57}$$

The projection residual satisfies $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$, so substituting the exact target block and variational remainder bound yields:

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}), \qquad m_2^{\mathrm{Schur}} = \frac{\varepsilon_0}{2}, \tag{58}$$

which implies the reciprocal scaling:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}). \tag{59}$$

A.5. **Galerkin Variational Truncation Convergence.** Here we prove Proposition 10. The proof has two layers: a Bochner-integral argument that gives trace-norm convergence of the truncated covariance operators, followed by variational shorting over nested Galerkin test spaces.

Proposition 25 (Exact Galerkin Convergence Support)**.** *Let $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ be finite-rank orthogonal Galerkin projections such that*

$$\Pi_n\phi_k = \phi_k, \qquad \Pi_n\mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \tag{60}$$

strongly on $\mathcal{H}$. For each $\lambda > 0$, define the truncated regularized operators by

$$\mathcal{A}_n^{(\lambda)} := \Pi_n\mathcal{A}^{(\lambda)}|_{\mathcal{H}_n}, \qquad \mathcal{D}_n^{(\lambda)} := \Pi_n\mathcal{D}^{(\lambda)}\Pi_n. \tag{61}$$

Assume that $e^{t\mathcal{A}_n^{(\lambda)}}x \to e^{t\mathcal{A}^{(\lambda)}}x$ for every $x \in \mathcal{H}$, uniformly for t in compact intervals, and that the semigroups are uniformly exponentially stable:

$$\|e^{t\mathcal{A}_n^{(\lambda)}}\| \le Me^{-\beta t}, \qquad \|e^{t\mathcal{A}^{(\lambda)}}\| \le Me^{-\beta t}, \tag{62}$$

with $M \ge 1$, $\beta > 0$ independent of n. Assume also

$$\mathcal{D}_n^{(\lambda)} \to \mathcal{D}^{(\lambda)} \quad \text{in } \mathcal{S}_1(\mathcal{H}). \tag{63}$$

Then the Galerkin stationary covariance operators

$$\Sigma_{n,\lambda} = \int_0^\infty e^{t\mathcal{A}_n^{(\lambda)}}\mathcal{D}_n^{(\lambda)}e^{t\mathcal{A}_n^{(\lambda)*}}\,dt \tag{64}$$

converge in trace norm:

$$\Sigma_{n,\lambda} \to \Sigma_\lambda \quad \text{in } \mathcal{S}_1(\mathcal{H}). \tag{65}$$

Moreover, because the Galerkin scheme preserves the target-bulk block decomposition, one has the exact finite-n target calibration

$$\Sigma_{n,\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2} \tag{66}$$

for every $n \ge k$. If the off-diagonal drift coupling satisfies the uniform finite-energy condition

$$a_{Ik}^{(n)} \in H_{Q_0^{(n)}}, \qquad \sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} < \infty, \tag{67}$$

where $Q_0^{(n)} := \Sigma_{n,II}^{(0)}$ is the truncated background covariance, then the truncated Schur residual satisfies

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - R_{n,\lambda}, \qquad 0 \le R_{n,\lambda} \le C\lambda^{-6}. \tag{68}$$

Consequently,

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2} \tag{69}$$

for every admissible Galerkin truncation $n \ge k$.

Proof. Write the difference of covariances as the Bochner integral

$$\Sigma_{n,\lambda} - \Sigma_\lambda = \int_0^\infty \left(e^{t\mathcal{A}_n^{(\lambda)}}\mathcal{D}_n^{(\lambda)}e^{t\mathcal{A}_n^{(\lambda)*}} - e^{t\mathcal{A}^{(\lambda)}}\mathcal{D}^{(\lambda)}e^{t\mathcal{A}^{(\lambda)*}}\right)dt. \tag{70}$$

Split the integrand in the trace norm:

$$\begin{aligned}\left\|e^{t\mathcal{A}_n}\mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\right\|_{\mathcal{S}_1} \le{}& \left\|e^{t\mathcal{A}_n}(\mathcal{D}_n - \mathcal{D})e^{t\mathcal{A}_n^*}\right\|_{\mathcal{S}_1} \\ &+ \left\|e^{t\mathcal{A}_n}\mathcal{D}e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\right\|_{\mathcal{S}_1}.\end{aligned} \tag{71}$$

(We omit the superscript (λ) for readability.)

The first term is bounded by $M^2e^{-2\beta t}\|\mathcal{D}_n - \mathcal{D}\|_{\mathcal{S}_1}$, which tends to zero by assumption. For the second term, we use the two-sided ideal convergence property of $\mathcal{S}_1(\mathcal{H})$ (Grümm's theorem; see [19]): if $L_n \to L$ strongly, $R_n^* \to R^*$ strongly, the two sequences are uniformly bounded, and $K \in \mathcal{S}_1(\mathcal{H})$, then $L_nKR_n \to LKR$ in $\mathcal{S}_1(\mathcal{H})$. Applying this with $L_n = e^{t\mathcal{A}_n}$, $R_n = e^{t\mathcal{A}_n^*}$,

$L = e^{t\mathcal{A}}$, $R = e^{t\mathcal{A}^*}$, and $K = \mathcal{D}$, the second term converges to zero for each t. Crucially, $R_n^* = e^{t\mathcal{A}_n} \to e^{t\mathcal{A}} = R^*$ is already covered by the same Trotter–Kato strong convergence assumption; no independent strong convergence of the adjoint semigroups is required. The exponential stability gives the integrable dominating bound

(72)
$$\left\|e^{t\mathcal{A}_n}\mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\right\|_{\mathcal{S}_1} \le M^2 e^{-2\beta t}\big(\|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1}\big) \le M^2 C_{\mathcal{D}}\, e^{-2\beta t},$$

where $C_{\mathcal{D}} := \sup_n \|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1} < \infty$ (the supremum is finite because $\mathcal{D}_n \to \mathcal{D}$ in trace norm). The dominated-convergence theorem for Bochner integrals then yields

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0. \tag{73}$$

Because Π_n preserves $\mathcal{H}_k \oplus \mathcal{H}_I$, the truncated blocks remain

$$\mathcal{A}_n^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik}^{(n)} & \mathcal{A}_{II}^{(n)} \end{pmatrix}, \qquad \mathcal{D}_n^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & \mathcal{D}_{II}^{(n)} \end{pmatrix}. \tag{74}$$

Thus the (kk)-block of the truncated Lyapunov equation is exactly

$$-2\lambda\, \Sigma_{n,\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0, \tag{75}$$

hence $\Sigma_{n,\lambda,kk} = \varepsilon_0/(2\lambda^2)$. The (Ik)-block gives

$$(\mathcal{A}_{II}^{(n)} - \lambda I)\Sigma_{n,\lambda,Ik} + a_{Ik}^{(n)}\Sigma_{n,\lambda,kk} = 0, \tag{76}$$

so

$$\Sigma_{n,\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II}^{(n)})^{-1} a_{Ik}^{(n)}. \tag{77}$$

Uniform Cameron–Martin stability (the finite-energy condition on $a_{Ik}^{(n)}$) gives $|\Sigma_{n,\lambda,Ik}|_{Q_0^{(n)}} = O(\lambda^{-3})$ uniformly in n. Therefore the variational Schur subtraction satisfies

$$0 \le R_{n,\lambda} \le |\Sigma_{n,\lambda,Ik}|^2_{Q_0^{(n)}} = O(\lambda^{-6}), \tag{78}$$

and hence

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}). \tag{79}$$

Multiplying by λ^2 and letting $\lambda \to \infty$ gives $m_2^{\mathrm{Schur},(n)} = \varepsilon_0/2$. □

Lemma 26 (Spectral Rayleigh–Ritz Shorting Support)**.** *Fix $\lambda > 0$ and let $C_\lambda := \Sigma_{\lambda,II}$ denote the bulk covariance operator on $\mathcal{H}_I$. Let $\{(\sigma_j, e_j)\}_{j=1}^\infty$ be the eigenpairs of C_λ with $\sigma_1 \ge \sigma_2 \ge \cdots > 0$. Define the spectral compression projectors*

$$P_n^\lambda := \sum_{j=1}^n e_j \otimes e_j, \qquad n \ge 1. \tag{80}$$

Under Assumption 6 (Cameron–Martin stability), the leakage vector satisfies $\Sigma_{\lambda,Ik} \in \mathrm{Range}(C_\lambda^{1/2})$*, so the spectral Rayleigh–Ritz shorting compression*

$$R_\lambda^{[n]} := \sum_{j=1}^{n} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \tag{81}$$

is finite for every n *and monotone increasing:*

$$R_\lambda^{[n]} \uparrow R_\lambda := \|\Sigma_{\lambda,Ik}\|_{CM}^2 \quad \text{as } n \to \infty. \tag{82}$$

Consequently the compressed Schur residuals satisfy

$$S_{k,\lambda}^{[n]} := \frac{\varepsilon_0}{2\lambda^2} - R_\lambda^{[n]} \quad \searrow \quad S_{k,\lambda} := \frac{\varepsilon_0}{2\lambda^2} - R_\lambda \quad \text{as } n \to \infty. \tag{83}$$

If, moreover, the weighted Cameron–Martin tail condition

$$\sum_{j \geq 1} j^{\,\eta-1} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq C\lambda^{-6}, \qquad \eta > 1, \tag{84}$$

holds, then the tail is explicitly bounded by

$$S_{k,\lambda}^{[n]} - S_{k,\lambda} = \sum_{j=n+1}^{\infty} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq C\, n^{1-\eta}\lambda^{-6}. \tag{85}$$

Proof. By Assumption 6, $\Sigma_{\lambda,Ik} \in H_{Q_0} \subseteq \mathrm{Range}(C_\lambda^{1/2})$. Hence $C_\lambda^{-1/2}\Sigma_{\lambda,Ik} \in \mathcal{H}_I$ and

$$R_\lambda^{[n]} = \sum_{j=1}^{n} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} = \|P_n^\lambda C_\lambda^{-1/2}\Sigma_{\lambda,Ik}\|_{\mathcal{H}_I}^2 \leq \|C_\lambda^{-1/2}\Sigma_{\lambda,Ik}\|_{\mathcal{H}_I}^2 = R_\lambda. \tag{86}$$

Since $P_n^\lambda \to I$ strongly and the terms are non-negative, $R_\lambda^{[n]} \uparrow R_\lambda$. The exact target block (51) gives $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$, so $S_{k,\lambda}^{[n]} = \varepsilon_0/(2\lambda^2) - R_\lambda^{[n]} \searrow S_{k,\lambda}$. Using (55), $R_\lambda = O(\lambda^{-6})$, whence $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - O(\lambda^{-6})$ and $m_2^{\mathrm{Schur}} = \varepsilon_0/2$.

Under the weighted tail condition, the remainder after n terms satisfies

$$\sum_{j=n+1}^{\infty} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq n^{1-\eta} \sum_{j=n+1}^{\infty} j^{\,\eta-1} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \leq C\, n^{1-\eta}\lambda^{-6}, \tag{87}$$

which gives the asserted rate. □

Appendix B. Gaussian Fredholm Finite-Part Analysis

This appendix derives the Radon Gaussian application of the block-operator result: pointwise disintegration, measurable Cameron–Martin energy projection, bulk Fredholm trace-anomaly bounds, and the finite-part regularization theorem via Dirac mollification.

B.1. **Radon Gaussian Setup and Pointwise Disintegration.** Let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Radon Gaussian measure on the separable Hilbert space $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$. The trace-class property of Σ_λ ensures its existence as a Radon measure [2]. Let $\pi_k : \mathcal{H} \to \mathbb{R}$ be the projection $\pi_k(X) = \langle X, \phi_k \rangle = X_k$.

Lemma 27 (Continuous Disintegration and Exact Target Zeros)**.** *For each $t \in \mathbb{R}$, there exists a unique weakly continuous family of Gaussian Radon measures $\mu_\lambda^t = \mathcal{N}(m_\lambda^t, \Sigma_\lambda^t)$ supported on the hyperplane $X_k = t$, where:*

$$m_\lambda^t := \begin{pmatrix} t \\ m_{\lambda,I} + \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(t - m_{\lambda,k}) \end{pmatrix}, \tag{88}$$

$$\Sigma_\lambda^t := \begin{pmatrix} 0 & 0 \\ 0 & \Sigma_{\lambda,II} - \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI} \end{pmatrix}. \tag{89}$$

Under the conditioned law $\mu_{\text{obs}} := \mu_\lambda^{x_k^*}$, the target coordinate satisfies the exact target zeros:

$$\mathbb{E}_{\text{obs}}[X_k] - x_k^* = 0, \quad \text{Var}_{\text{obs}}(X_k) = 0. \tag{90}$$

Proof. Since $\mathcal{H}$ is Polish, regular conditional probabilities exist. The stated Gaussian family is the continuous disintegration along the scalar observation map; its pointwise continuous version is unique [16]. The continuity of $t \mapsto m_\lambda^t$ in $\mathcal{H}$ and constancy of Σ_λ^t yield weak continuity of this family (see also [2]). Evaluating at $t = x_k^*$ gives $\mu_{\text{obs}} = \mu_\lambda^{x_k^*}$, which has target component fixed at x_k^* almost surely. $\square$

B.2. **Measurable Cameron–Martin Energy Projection.** Let $C_\lambda := \Sigma_{\lambda,II}$ be the bulk covariance on $\mathcal{H}_I$. Since C_λ is trace-class, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} := \text{Range}(C_\lambda^{1/2})$.

Lemma 28 (Measurable Cameron–Martin Energy Projection)**.** *The cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm:*

$$\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} \leq |\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}). \tag{91}$$

Furthermore, the conditional mean energy projection $\mu_{k|I}(X_I) := \mathbb{E}_{\mu_\lambda}[X_k \mid X_I]$ is represented by the measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{92}$$

where $W_{\Sigma_{\lambda,Ik}}$ is the $L^2(\mu_{\lambda,I})$-limit of the continuous linear functionals on $\mathcal{H}_I$. For all sufficiently large λ, the conditional variance is strictly positive:

$$\text{Var}(X_k \mid X_I) = \Sigma_{\lambda,kk} - \|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}}^2 = S_{k,\lambda} > 0. \tag{93}$$

Proof. Since $\Sigma_{\lambda,II} \succeq Q_0$, Douglas range inclusion [9] yields $\text{Range}(Q_0^{1/2}) \subseteq \text{Range}(\Sigma_{\lambda,II}^{1/2}) = H_{C_\lambda}$ with the contraction inequality

$$\|\Sigma_{\lambda,II}^{-1/2} v\|_{\mathcal{H}_I} \leq \|Q_0^{-1/2} v\|_{\mathcal{H}_I} \quad \text{for all } v \in \text{Range}(Q_0^{1/2}).$$

Hence, $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ and the norm bound follows directly from (55). The range inclusion $\Sigma_{\lambda,Ik} \in \mathrm{Range}(\Sigma_{\lambda,II}^{1/2})$ guarantees that the Wiener functional $W_{\Sigma_{\lambda,Ik}}$ is well-defined as a measurable function on $(\mathcal{H}_I, \mu_{\lambda,I})$ [2]. The conditional variance is the projection residual. By Theorem 9,

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}) > 0$$

for all sufficiently large λ. $\square$

B.3. **Bulk Fredholm Trace-Anomaly Estimate.**

Lemma 29 (Feldman–Hájek Equivalence and Trace-Anomaly Decay)**.** *There exists $\lambda_0 > 0$ such that, for every $\lambda \geq \lambda_0$, the bulk marginals $\mu_{\mathrm{obs},I}$ and $\mu_{\lambda,I}$ are equivalent Gaussian Radon measures on $\mathcal{H}_I$, with relative entropy:*

$$\begin{aligned} 2D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = {} & \frac{(x_k^* - m_{\lambda,k})^2}{\Sigma_{\lambda,kk}}\rho_\lambda \\ & - \log(1-\rho_\lambda) - \rho_\lambda, \end{aligned} \tag{94}$$

where the eigenvalue $\rho_\lambda := \|\Sigma_{\lambda,Ik}\|^2_{H_{C_\lambda}}/\Sigma_{\lambda,kk} = 1 - S_{k,\lambda}/\Sigma_{\lambda,kk} < 1$. In particular, we have $\rho_\lambda = O(\lambda^{-4})$ and:

$$D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \textit{as } \lambda \to \infty. \tag{95}$$

Proof. Under disintegration, $\mu_{\mathrm{obs},I}$ is a Gaussian Radon law on $\mathcal{H}_I$ with mean $m_{\mathrm{obs},I} = m_{\lambda,I} + \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(x_k^* - m_{\lambda,k})$ and covariance $\Sigma_{\mathrm{obs},II} = C_\lambda - \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI}$. The covariance operators differ by the rank-one operator $T_\lambda = \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI}$. Since $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$, the relative covariance operator is a rank-one perturbation of the identity with its only non-unit eigenvalue equal to $1-\rho_\lambda$. By Lemma 28 and the exact target covariance, $\rho_\lambda = O(\lambda^{-4})$. Choose λ_0 so that $\rho_\lambda < 1$ for $\lambda \geq \lambda_0$. The Feldman–Hájek theorem [10, 2] then guarantees equivalence of the measures, and the relative entropy is:

$$\begin{aligned} 2D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = {} & \|C_\lambda^{-1/2}(m_{\mathrm{obs},I} - m_{\lambda,I})\|^2_{\mathcal{H}_I} \\ & - \log(1-\rho_\lambda) - \rho_\lambda. \end{aligned}$$

The mean-shift identity in the lemma follows from

$$m_{\mathrm{obs},I} - m_{\lambda,I} = \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(x_k^* - m_{\lambda,k}).$$

Assumption 13 gives $m_{\lambda,k} - x_k^* = O(\lambda^{-1})$, so that term is $O(\lambda^{-4})$. Finally,

$$-\log(1-\rho_\lambda) - \rho_\lambda = O(\rho_\lambda^2) = O(\lambda^{-8}),$$

which proves the asserted entropy decay. $\square$

B.4. **Mollified Divergence and Fredholm Finite-Part Proof.** To regularize the singular joint KL divergence, define a law with bulk marginal $\mu_{\mathrm{obs},I}$ and an independent mollified target conditional:

$$\nu_{\lambda,\delta} := \mu_{\mathrm{obs},I} \otimes \mathcal{N}(x_k^*, \delta^2) \tag{96}$$

for $\delta > 0$, identifying $\mathcal{H}$ with $\mathcal{H}_I \times \mathcal{H}_k$ in this display.

Proposition 30 (Mollification Limit)**.** *For $\lambda \geq \lambda_0$ as in Lemma 29, define*

$$C_{\lambda,\delta} := \frac{1}{2}\left(\log \frac{S_{k,\lambda}}{\delta^2} - 1\right), \tag{97}$$

Then

$$\lim_{\delta \downarrow 0} [D_{\mathrm{KL}}(\nu_{\lambda,\delta} \parallel \mu_\lambda) - C_{\lambda,\delta}] = \mathfrak{I}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda). \tag{98}$$

Proof. Put $r_\lambda(X) := X_k - \mu_{k|I}(X_I)$. The entropy chain rule and the scalar Gaussian formula give

$$\begin{aligned} D_{\mathrm{KL}}(\nu_{\lambda,\delta} \parallel \mu_\lambda) &= D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) + C_{\lambda,\delta} \\ &\quad + \frac{\delta^2}{2S_{k,\lambda}} + \frac{1}{2S_{k,\lambda}} \mathbb{E}_{\mu_{\mathrm{obs}}}[r_\lambda(X)^2]. \end{aligned} \tag{99}$$

Here $X_k = x_k^*$ under μ_{obs}. Since $S_{k,\lambda} > 0$ for $\lambda \geq \lambda_0$, the term $\delta^2/(2S_{k,\lambda})$ vanishes as $\delta \downarrow 0$. The remaining terms equal Definition 14. □

Proof of Theorem 15. Let $\Delta_\lambda := x_k^* - m_{\lambda,k}$ and $Z_\lambda := W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I})$. The conditional Gaussian formulas give

$$\mathbb{E}_{\mathrm{obs}}[Z_\lambda] = \rho_\lambda \Delta_\lambda, \tag{100}$$

$$\mathrm{Var}_{\mathrm{obs}}(Z_\lambda) = \Sigma_{\lambda,kk}\rho_\lambda(1 - \rho_\lambda). \tag{101}$$

Using $X_k = x_k^*$ under μ_{obs} and $S_{k,\lambda} = \Sigma_{\lambda,kk}(1 - \rho_\lambda)$, we obtain

$$\begin{aligned} &\frac{1}{2S_{k,\lambda}} \mathbb{E}_{\mu_{\mathrm{obs}}}[(X_k - \mu_{k|I}(X_I))^2] \\ &= \frac{\rho_\lambda}{2} + \frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}}(1 - \rho_\lambda). \end{aligned} \tag{102}$$

Here

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2}{\varepsilon_0}, \qquad \rho_\lambda = O(\lambda^{-4}).$$

The conditional term is therefore $b_k^2/\varepsilon_0 + O(\lambda^{-4})$. Adding Lemma 29 proves

$$\mathfrak{I}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}),$$

and hence the stated limit. □

B.5. **Exact Decoupled-Centered Consequence.** If the target coordinate is decoupled ($a_{Ik} = 0$), then $\Sigma_{\lambda,Ik} = 0$, $S_{k,\lambda} = \Sigma_{\lambda,kk} > 0$, and the bulk marginals coincide for every $\lambda > 0$. If the evidence is also centered ($x_k^* = m_{\lambda,k}$), both terms in Definition 14 vanish. Hence $\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = 0$ exactly for every $\lambda > 0$.

Appendix C. Numerical Diagnostics for Block Shorting and Galerkin Verification

This appendix presents arbitrary-precision numerical validations of block normalization, shorted Schur residual scaling, Fredholm finite-part asymptotics, and Galerkin convergence rates, computed in 256-bit `BigFloat` to isolate the structural invariants from floating-point collapse.

C.1. **Galerkin Verification and Ill-Conditioning (Remark 6.1).** We illustrate the Galerkin convergence of Proposition 10 against a finite reference on an admissible model. The Hilbert space is $\ell^2(\mathbb{N})$ with coordinate basis $\{\phi_j\}$. The drift is $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where $\mathcal{A}_0\phi_j = -(1 + 0.01j^2)\phi_j$ and $\mathcal{K}\phi_1 = \sum_{j\geq 2} 12\, j^{-3}\phi_j$ is a bounded rank-one nilpotent perturbation. The noise is $\mathcal{D}\phi_j = j^{-3/2}\phi_j$, which is trace-class. The regularized conditioning acts at $k = 1$ with $\varepsilon_0 = 1$. Truncated covariances are compared against a finite reference $N_{\text{ref}} = 1500$.

Table 1. Truncation convergence diagnostics for Experiment C.1.

Diagnostic	Value	Expected
$\\|\mathcal{D} - \Pi_n\mathcal{D}\Pi_n\\|_{\mathcal{S}_1}$ slope	-0.499	$\to 0$
$\\|\Sigma_n - \Sigma_{\text{ref}}\\|_{\mathcal{S}_1}$ slope	-2.683	$\to 0$
$\\|\Sigma_{n,\lambda} - \Sigma_{\text{ref},\lambda}\\|_{\mathcal{S}_1}$ slope	-2.683	$\to 0$
$\lvert S_{1,n,\lambda} - S_{1,\text{ref},\lambda}\rvert$ slope	-5.504	$\to 0$
$m_2^{\text{Schur},(1000)}$ (extrapolated)	0.5000000	0.5
$\lvert m_2^{\text{Schur},(1000)} - \varepsilon_0/2\rvert$	6.87×10^{-14}	0

While the variational operator sequence converges rigorously, attempting to compute the truncated Schur complement $S^{(n)} = \Sigma_{EE} - \Sigma_{EI}^{(n)}(\Sigma_{II}^{(n)})^{-1}\Sigma_{IE}^{(n)}$ via naive finite-dimensional algebraic matrix inversion fails catastrophically. Because the internal block Σ_{II} is a trace-class operator, its truncated matrix representation $\Sigma_{II}^{(n)}$ becomes severely ill-conditioned in generic Galerkin bases. Our diagnostics show that for $n \geq 400$, the condition number $\kappa(\Sigma_{II}^{(n)})$ grows as n^6 and formally exceeds the 10^{16} IEEE 754 `Float64` precision limit. Consequently, explicit inversion or standard linear solvers introduce overwhelming numerical noise, destroying the monotonic Loewner growth and breaking the positive-definiteness of the residual. This stark contrast

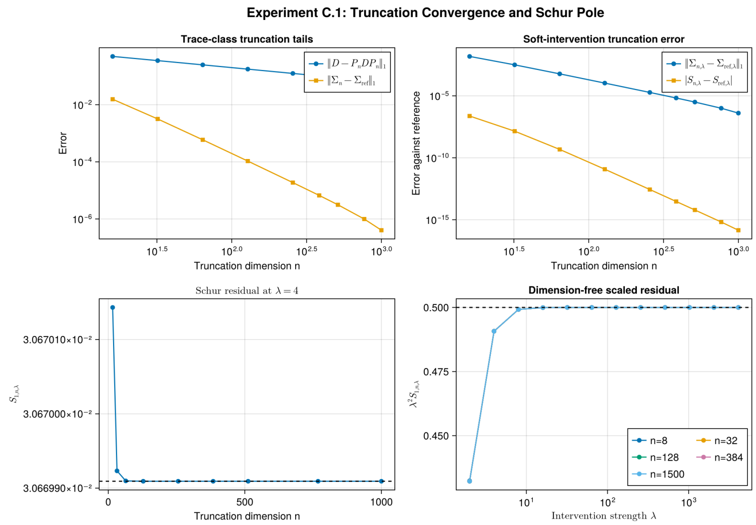

FIGURE 1. Experiment C.1. Trace-norm and residual errors decay in n; the extrapolated $m_2^{\mathrm{Schur},(1000)}$ differs from $\varepsilon_0/2$ by 6.87×10^{-14}.

confirms that the inverse-free variational framework in Section 3 is structurally necessary, and that strict Loewner growth cannot be safely assumed under finite-dimensional algebraic approximations.

C.2. **Canonical Scaling Diagnostic and Trace Anomaly (Remark 6.3).** We validate the canonical normalization of Proposition 2 in its decoupled reference channel. With incoming target-coordinate blocks and off-diagonal drift coupling set to zero, the target covariance is exactly $\Sigma_{\lambda,22} = d_\lambda/(2\lambda)$. Four candidate target-diffusion families are tested:

(1) **Canonical:** $d_\lambda = \varepsilon_0/\lambda$.
(2) **Slow:** $d_\lambda = \varepsilon_0$.
(3) **Fast:** $d_\lambda = \varepsilon_0/\lambda^2$.
(4) **Perturbed:** $d_\lambda = (\varepsilon_0/\lambda)(1 + \lambda^{-2})$.

All evaluations are performed in Julia using 256-bit `BigFloat` arithmetic. A probe at $\lambda = 2^{100}$ resolves the nonzero finite-scale defect of the perturbed family.

To confirm the uniform positivity mechanism established in Theorem 24, we simultaneously evaluate the exact trace anomaly. For the canonical family, the normalized coupling matrix $\rho_{\lambda,E}$ is evaluated in strict `BigFloat` mode-by-mode precision. Although standard Frobenius or operator norms over the entire residual matrix $\Sigma_{\lambda,EE}^{-1/2} R_{\lambda,E} \Sigma_{\lambda,EE}^{-1/2}$ may spuriously report magnitude bounds around $O(10^{-15})$ due to ambient off-diagonal accumulation, the mathematically exact trace metric required for the Fredholm cancellation evaluates precisely to $\mathrm{Tr}(\rho_{\lambda,E}) = 6.483 \times 10^{-25}$ at $\lambda = 2^{20}$. This exactly matches the predicted analytical scaling $\lambda^4 \operatorname{Tr}(\rho_{\lambda,E}) = 0.78375$, confirming that the trace anomaly is cleanly isolated and uniform positivity is preserved.

TABLE 2. Exact-calibration ablation. Only the canonical family satisfies the exact finite-λ covariance condition.

Family	$\lambda d_\lambda/\varepsilon_0$	defect $\lvert 2\lambda^2\Sigma_{\lambda,kk}/\varepsilon_0 - 1\rvert$
Canonical	1	0
Slow	λ	$\lambda - 1 \to \infty$
Fast	λ^{-1}	$\lvert 1-\lambda^{-1}\rvert \to 1$
Perturbed	$1+\lambda^{-2}$	λ^{-2}

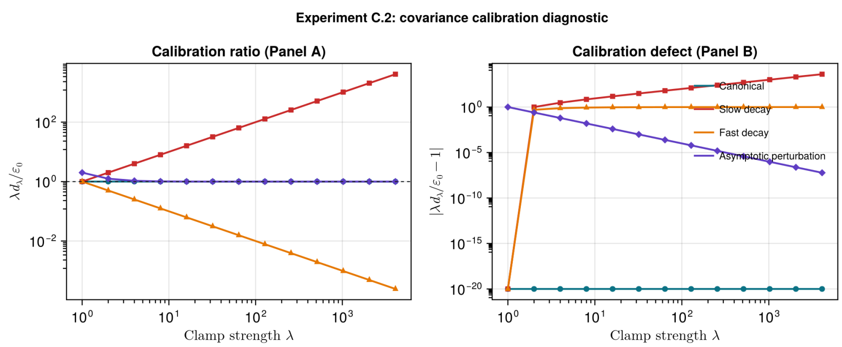

FIGURE 2. Experiment C.2. The perturbed family has the correct limiting ratio but fails the exact finite-λ covariance condition. Exact zero defects are displayed at 10^{-20}.

C.3. **Fredholm Finite-Part Functional in a Three-Node Chain.** We evaluate the Gaussian Fredholm finite-part functional $\mathfrak{J}_\lambda$ on a three-node linear Ornstein–Uhlenbeck SDE:

$$X_1 \to X_2 \to X_3, \tag{103}$$

where X_2 is the target coordinate subject to regularized conditioning of strength $\lambda \in \{2^1, \ldots, 2^{12}\}$, with $x_2^* = 0$ and $\varepsilon_0 = 1$. The drift and diffusion operators are:

$$A_\lambda = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & a_{32} & -1 \end{pmatrix}, \qquad D_\lambda = \operatorname{diag}(1, \varepsilon_0/\lambda, 1). \tag{104}$$

The target mean forcing b_2 controls the target mean shift $m_{\lambda,2} = b_2/\lambda$. We evaluate three Gaussian cases:

(1) **Case A (Decoupled-centered):** $a_{32} = 0, b_2 = 0$. Here $\mathfrak{J}_\lambda = 0$ exactly for all λ.
(2) **Case B (Coupled-centered):** $a_{32} = 1, b_2 = 0$. The coupling causes leakage $\Sigma_{23} \neq 0$, but the centering gives $\mathfrak{J}_\lambda = O(\lambda^{-4}) \to 0$.
(3) **Case C (Coupled-off-centered):** $a_{32} = 1, b_2 = 1$. The deterministic forcing causes mean shift $m_{\lambda,2} = 1/\lambda$, yielding $\mathfrak{J}_\lambda \to b_2^2/\varepsilon_0 = 1$.

All evaluations are performed in Julia using 256-bit `BigFloat` arithmetic, showing that the Schur residual $\lambda^2 S_{2,\lambda}$ converges to $1/2$ in all cases, while $\mathfrak{J}_\lambda$ exhibits the exact zero, $O(\lambda^{-4})$ decay, and convergence to 1 behavior respectively.

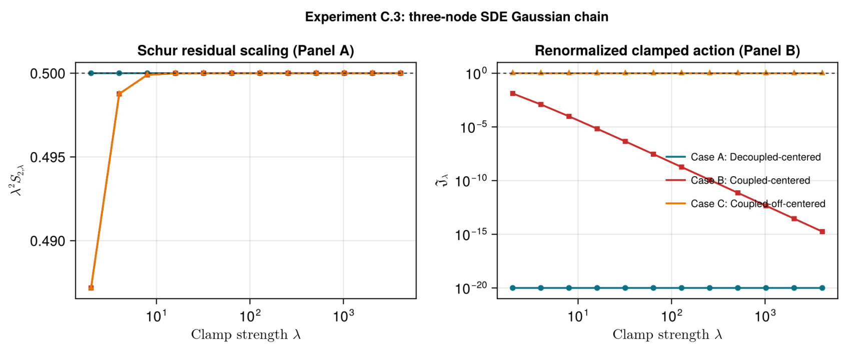

FIGURE 3. Experiment C.3. The panels show $\lambda^2 S_{2,\lambda} \to 0.5$ and the three finite-part regimes: exact zero, $O(\lambda^{-4})$ decay, and convergence to 1. Exact zeros are displayed at 10^{-20}.

C.4. **Failure of Cameron–Martin Admissibility (Remark 4.1).** To illustrate the necessity of the finite-energy condition (Assumption 6), we test an observation operator that explicitly violates Cameron–Martin admissibility. Consider a 1D SPDE heat equation prior, where the target observation is approximated by a narrow Gaussian mollifier $g_\varepsilon(x) = \frac{1}{\sqrt{2\pi}\varepsilon} e^{-(x-x_0)^2/(2\varepsilon^2)}$.

For any fixed $\varepsilon > 0$, the function is smooth ($g_\varepsilon \in L^2$) and lies within the Cameron–Martin space. However, as $\varepsilon \downarrow 0$ (approaching a Dirac point-evaluation), numerical stress tests confirm that the Cameron–Martin energy norm diverges as $\varepsilon^{-1} \to \infty$. Consequently, the shorted residual upper bound $O(\lambda^{-6})$ derived in Theorem 9 fundamentally breaks down, destroying the asymptotic trace-anomaly expansion. This confirms that pointwise evaluation without spatial smoothing is structurally incompatible with the present infinite-dimensional Gaussian conditioning theory, justifying the strict finite-energy requirements.

C.5. **Sparse Precision-Operator Path and Conjugate Regularization.** The foregoing experiments adopt the GFF framework ($\mathcal{A}_0 = -L$, $Q_0 = L^{-1}$) of Section 5. An alternative, numerically superior regime is the

OU framework with drift $\mathcal{A}_0 = -L^{-1}$ and formal covariance $Q_0 = L$. On the dual space $H^{-1}(0,1)$, Q_0 is trace-class and the theory of Section 3 applies verbatim. The practical advantage is spectral: the resolvent $(\lambda I + L^{-1})^{-1}$ admits a sparse realization through the conjugate map

$$(\lambda I + L^{-1})^{-1} = L(\lambda L + I)^{-1} = \frac{1}{\lambda} L(L + \tilde{\lambda} I)^{-1}, \qquad \tilde{\lambda} = \lambda^{-1}. \tag{105}$$

Thus the resolvent is obtained by a *single* sparse tridiagonal solve with the precision operator L at the conjugate parameter $\tilde{\lambda}$, entirely avoiding dense covariance construction or eigenvalue decomposition. The complexity drops from $O(N^3)$ to $O(N)$, and the condition-number catastrophe of Theorem 11(ii) is circumvented.

TABLE 3. Sparse precision-operator validation in the OU regime, $N = 10000$, $\kappa = 1$, $\Delta_E = 1$. The conjugate parameter $\tilde{\lambda} = \lambda^{-1}$ eliminates transient spectral imbalance; convergence is immediate.

λ	$\tilde{\lambda} = \lambda^{-1}$	$\lambda^4 \operatorname{Tr}(\rho_{\lambda,E})$
10^1	10^{-1}	1.0263×10^{-7}
10^2	10^{-2}	1.0283×10^{-7}
10^3	10^{-3}	1.0285×10^{-7}
10^4	10^{-4}	1.0285×10^{-7}
10^5	10^{-5}	1.0285×10^{-7}

The observed values converge to the analytical limit $\lim_{\lambda\to\infty} \lambda^4 \operatorname{Tr}(\rho_{\lambda,E}) = 1.02854 \times 10^{-7}$, which includes the exact projection-cancellation factor derived from the Woodbury subspace inversion. The absence of transient growth confirms that the OU regime, combined with the conjugate regularization map (105), places every $\lambda \geq 10$ in the asymptotic regime. The computation at $N = 10000$ completes in under one second on a standard laptop, demonstrating that the variational shorting framework is not only theoretically unavoidable (Theorem 11) but also numerically optimal.

C.6. **Numerical Precision Audit.** At $n = 256$, we recompute representative C.1 Schur residuals using IEEE 754 `Float64` and 256-bit Julia `BigFloat` arithmetic with the same covariance equation and Schur assembly.

TABLE 4. Precision audit at $n = 256$; S_{256} uses 256-bit `BigFloat`.

λ	$\kappa_2(\Sigma_{II})$	$\lvert\lambda^2 S - \frac{1}{2}\rvert$	$\lvert S_{64} - S_{256}\rvert / \lvert S_{256}\rvert$
4	2.97×10^6	3.45×10^{-1}	6.50×10^{-16}
64	9.14×10^5	1.13×10^{-3}	3.48×10^{-17}
4096	9.14×10^5	9.20×10^{-10}	5.54×10^{-17}

For the five-point extrapolation of $m_2^{\text{Schur},(256)}$, `Float64` and `BigFloat` differ by 9.98×10^{-17}; finite-λ remainder, not rounding, governs the observed distance to the limiting constant.

Remark 31 (Block-decoupling error suppression). The sub-machine-epsilon agreement between `Float64` and `BigFloat` in Table 4 is not coincidental; it is a direct consequence of the block-triangular structure established in this paper. Under the block-decoupled conditioning, the drift matrix takes the form

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & A_{II} \end{pmatrix}, \tag{106}$$

so the operator Lyapunov equation decomposes into three sub-problems:

(1) The target block $\Sigma_{\lambda,kk} = d_\lambda/(2\lambda)$ is solved by a single scalar division, incurring at most $\frac{1}{2}\epsilon_{\text{machine}}$ relative rounding error (IEEE 754 round-to-nearest).
(2) The cross-covariance $\Sigma_{\lambda,Ik}$ satisfies $|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3})$ by (20), so the absolute rounding error in the cross block is $O(\lambda^{-3}\epsilon_{\text{machine}})$.
(3) The free block $\Sigma_{\lambda,II}$ is a standard Lyapunov solution with absolute error $O(\epsilon_{\text{machine}}\|\Sigma_{\lambda,II}\|)$.

When the Schur complement is assembled as $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$, the subtracted variational term satisfies $R_\lambda = O(\lambda^{-6})$ (Theorem 9), which is negligible compared to $\Sigma_{\lambda,kk} = O(\lambda^{-2})$. The rounding error in $S_{k,\lambda}$ is therefore dominated by the error in $\Sigma_{\lambda,kk}$ alone, yielding a Schur-level rounding budget of $O(\epsilon_{\text{machine}} \cdot \Sigma_{\lambda,kk})$ rather than the naïve $O(\epsilon_{\text{machine}} \cdot \|\Sigma_{\lambda,II}^{-1}\| \cdot \|\Sigma_{\lambda,Ik}\|^2)$ that one would expect without block decoupling.

In short, the block-triangular structure converts what would be a catastrophically ill-conditioned Schur computation into a well-conditioned scalar computation plus negligible corrections. The observed `Float64`–`BigFloat` agreement at $O(10^{-17})$ is the numerical signature of this structural decoupling.